An Unambiguous Imaging Method of Moving Target for Maritime Scenes With Spaceborne High-Resolution and Wide-Swath SAR

Yajun Long, Fengjun Zhao, Mingjie Zheng, Liangbo Zhao, and Wenjing He

***Abstract*— In azimuth multichannel high-resolution and wide-swath (AMC-HRWS) synthetic aperture radar (SAR) system, undersampling of azimuth signal results in the failure of the single-channel echo imaging, so the echoes of all the channels should be reconstructed to obtain the unambiguous SAR image. Many methods have been proposed to reconstruct the stationary scene echoes. However, moving targets (MTs) will cause false targets in the reconstructed SAR image, which has a great influence on SAR image interpretation and moving target detection, especially in maritime scenes. Hence, we propose an unambiguous imaging method of the moving target for maritime scenes with AMC-HRWS SAR. First, azimuth deramp processing is introduced to obtain coarse-focused moving targets, and then their echoes can be extracted from the sea clutter. Based on the extracted data, a radial velocity estimation method is proposed, which is high efficiency and does not require redundant channels. In addition, based on the estimated radial velocity and the characteristics of the coarse-focused signal, a novel signal reconstruction method of moving target is presented. The reconstruction performance of this method is not affected by channel imbalance, and false targets can be removed even if there is radial velocity error. Besides, the effectiveness of this scheme is verified by the simulation data and GaoFen-3 SAR real data. The estimated radial velocities of ships are verified by automatic identification system (AIS) information. The imaging results show that false targets are effectively suppressed, and moving targets are located in correct positions.**

***Index Terms*— Azimuth multichannel (AMC), high-resolution and wide-swath (HRWS), maritime scene, moving target (MT) signal reconstruction, radial velocity estimation, synthetic aperture radar (SAR).**

I. Introduction

DUE to the limitation of pulse repetition frequency (PRF) selection, the conventional synthetic aperture radar (SAR) system cannot meet the increasing demands of high-resolution and wide-swath (HRWS) imaging simultaneously. High azimuth resolution requires a high PRF, whereas a lower PRF is required to avoid range ambiguity [1]. Azimuth multichannel (AMC) SAR systems are equipped with multiple antennas along the track to receive the echoes from the same illuminated area. The increase in azimuth spatial sampling rate reduces the demand for time sampling rate, thereby decreasing the PRF of the system, which relaxes the inherent limitation of the single-channel SAR system and enables HRWS imaging [2]. The existing spaceborne SAR systems, such as TerraSAR-X [3], [4], ALOS-2 [5], RADARSAT-2 [6], and Chinese GaoFen-3 [7], which are equipped with dual-receiving channel (DRC), demonstrate the feasibility of the AMC-HRWS SAR imaging technique.

Ground moving target indication (GMTI) is one of the primary applications of the spaceborne HRWS SAR, especially in maritime scenes [8]. Moving target (MT) detection, motion parameter estimation, and relocation are the main tasks of GMTI. The classical multichannel GMTI techniques, such as along-track interferometry (ATI) [9], extended displaced phase center antenna (EDPCA) [10], [11], and space–time adaptive processing (STAP) [12], improve the detection performance of moving targets. However, these methods require that the PRF of each channel be greater than the azimuth bandwidth, so they are not suitable for AMC-HRWS systems. For AMC-HRWS SAR, the Doppler spectrum of each channel signal is aliased, and signal reconstruction of the moving target is also a crucial task of GMTI. Many signal reconstruction methods have been proposed for stationary scenes, such as the reconstruction filter method [13], [14] and the STAP-based method [15], [16]. Most of them essentially make use of the relationship between the instantaneous Doppler frequency and the cone angle to reconstruct the unambiguous Doppler spectrum, so as to obtain the unambiguous SAR image. Different from the static target (ST), the signal model of the moving target is related to its radial velocity and azimuth velocity [17]. The radial velocity introduces an offset of the Doppler spectrum [18], which makes the relationship between the instantaneous Doppler frequency and the cone angle be not determined. The aliased Doppler spectrum of the moving target cannot be correctly reconstructed together with that of the stationary scene, resulting in false targets (or named azimuth ambiguity) in the reconstructed SAR image. The Doppler spectrum offset also makes the focused position of

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the real moving target deviate from the true azimuth position. In addition, the linear range cell migration (RCM) caused by the radial velocity results in range and azimuth defocus of the moving target. Thus, radial velocity estimation is an important step to realize unambiguous imaging and relocation of the moving target.

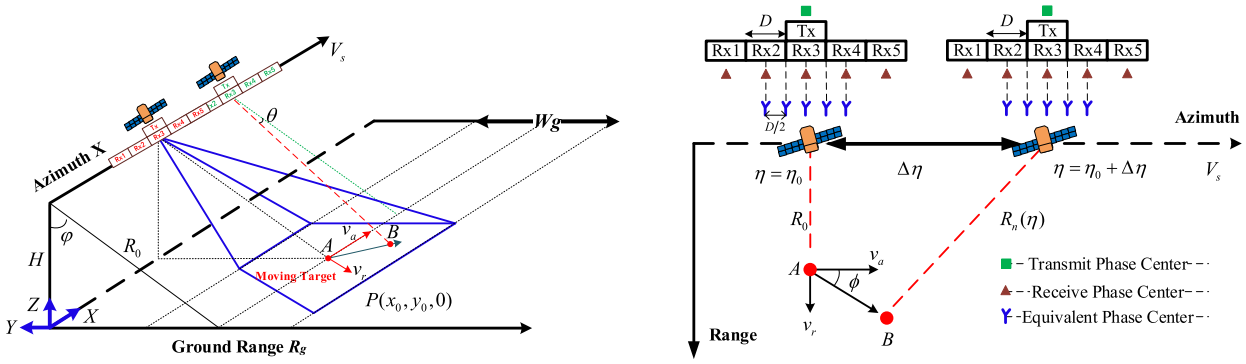
To reconstruct the moving target signal, some methods of radial velocity estimation and signal reconstruction have been proposed. [19], [20], and [21] proposed the reconstruction method based on the idea of reconstruction filter. In method [19] and [20], different radial velocities are taken into reconstruction filter and imaging flow to obtain the best matching radial velocity by maximizing the signal-to-clutter ratio (SCR) of the real target and obtain the reconstructed target simultaneously. A radial velocity estimation strategy based on signal subspace [22] is also processed in an iterative way with different velocity assumptions. A method based on maximum likelihood minimum entropy is proposed in [23], in which a phase mismatch term is used to estimate the radial velocity. The phase mismatch value is obtained by minimizing the entropy through multiple iterations. In addition to the heavy computation caused by iteration, a redundant channel is needed in method [22], and the uniform sampling PRF is required in method [23]. The method proposed in [24] estimates the radial velocity based on the energy distribution of the azimuth reconstructed signal in the Doppler frequency domain. However, only the result of the ideal point target is given, and the influence of clutter on radial velocity estimation is not considered. References [25] and [26] use the ATI-based method to estimate the radial velocity, whereas an additional image processing is needed to determine the position of the real moving target. The shift of Doppler spectrum makes the direction of arrival (DOA) of the moving target different from that of the ST. Yang *et al.* [27] propose a radial velocity estimation method based on DOA estimation. This method has high-resolution DOA estimation ability under high SCR, while the estimation accuracy of DOA cannot be guaranteed. Similarly, in [28], a method based on maximum likelihood (ML-based) is proposed to estimate the cone angle and radial velocity, but the redundant channel is also needed. The modified frequency correlation method (MFCM) is proposed in [29], which uses the correspondence between the RCM curve and Doppler spectrum to obtain the echo data near zero frequency by intercepting a small part in the middle of the RCM curve, and then the radial velocity is estimated by frequency correlation after the azimuth delay between channels is compensated. This method is free from iteration operation and high accuracy.

Since there are many moving ships in the maritime scene, and their motion parameters are different, moving targets needs to be processed separately. Among the above methods, [22] and [23] mentioned the echo extraction problem of moving target, both of which are performed in the range–Doppler (R-D) domain. Due to RCM, the data structure of the moving target occupies multiple range cells in the R-D domain so that the adjacent moving targets in the range direction are impossible to be separated. The azimuth deramp technique is derived from the Specan algorithm for SAR imaging [30], which can compress the Doppler spectrum of the signal and transform it into a single-frequency signal. For AMC-HRWS SAR, the azimuth deramp operation can be carried out on the echoes of all the channels to compress the aliased Doppler spectrum, so as to obtain the coarse-focused SAR images of each channel echo in the range time and azimuth frequency domain. The scheme proposed in [31] and [32] uses redundant channels to separate the echoes of stationary scene and moving target in coarse-focused image domain through beamforming technology. In [33], a simultaneous imaging scheme of sea clutter and moving targets for maritime scenes is proposed. The moving target echo is also extracted from the coarse-focused SAR images. In this scheme, the time-domain correlation (TDC) method and ML-based method [28] are used to estimate the radial velocity. The former obtains the velocity by estimating the interferometric phase. However, the azimuth delay between channels is difficult to compensate accurately in the case of undersampling, resulting in inaccurate calculation of interferometric phase. The phase error will accumulate with the increase in ambiguity number.

In this article, to achieve unambiguous imaging of moving target for maritime scenes with AMC-HRWS SAR, the novel methods of radial velocity estimation and signal reconstruction are proposed. The complete data processing flow is given, including moving target detection, echo extraction, radial velocity estimation, signal reconstruction, and imaging. First, range compression and azimuth deramp operation is performed on the echo data to obtain coarse-focused SAR images, and coarse-focused moving targets are detected by generalized likelihood ratio test (GLRT) algorithm [32], [34], and then their echoes are extracted from the sea clutter. With this operation, moving targets adjacent in the range direction can be separated. In the coarse-focused image domain, the characteristics of the moving target signal are analyzed, and thus a novel radial velocity estimation method is proposed. Thereinto, the radial velocity is divided into the baseband part and the ambiguity part, which are estimated by estimating the signal's spatial spectrum and minimizing the entropy of the coarse-focused moving target, respectively. In addition, based on the estimated radial velocity and the characteristics of the coarse-focused signal, a novel signal reconstruction method of moving target is presented. The reconstruction performance of this method is not affected by channel imbalance, and false targets can be removed even if there is radial velocity error. The proposed methods can be used not only for the existing DRC SAR systems but also for future multiple-channel HRWS SAR systems. Both multichannel simulation data and DRC real data of GaoFen-3 SAR are used to verify the effectiveness of the proposed methods. The estimated velocity velocities of ships are verified by automatic identification system (AIS) data. The proposed radial velocity method can achieve accurate and efficient velocity estimation without redundant channels. False targets can be effectively suppressed and moving targets can be relocated to correct positions through the proposed signal reconstruction method.

This article is arranged as follows. In Section II, the signal model of the moving target for the AMC-HRWS SAR system is described. The methods of moving target detection, echo

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(a) (b)

Fig. 1. Imaging geometry of the AMC-HRWS SAR system. (a) Imaging geometry for side-looking mode. (b) Distance history of the moving target in the slant-range plane, and the positions of EPCs.

extraction, radial velocity estimation, and signal reconstruction are presented in Section III. In Section IV, some problems about azimuth deramp processing are discussed. In Section V, the experimental results of the simulation data and the DRC data of GaoFen-3 SAR are presented. Conclusions are given in Section VI.

II. Signal Model of AMC-HRWS SAR

The imaging geometry of the AMC-HRWS SAR system for side-looking mode is shown in Fig. 1(a), where X-axis is the flight direction of the platform (azimuth), Y-axis points to the left (satisfying the orthogonal right-hand rule), and Z-axis is far away from the center of the earth. The radar platform velocity is V_s, and the height is H. W_g is the swath width on the ground, φ is the depression angle, and θ is the cone angle.

In general, the whole antenna is uniformly divided into N subapertures in azimuth. The chirp signal is transmitted by the middle subaperture with a wide beam, whereas the echoes from the same illuminated area are received by all subapertures simultaneously. Taking a five-channel system (i.e., $N = 5$) as an example, which is shown in Fig. 1(b). Tx and Rx are the transmitting and receiving channels, respectively, and ϕ is the angle between the target motion direction and the azimuth direction. Assuming that the distance between the phase centers of the adjacent receiving channels is D. According to the derivation in [14], by compensating a constant phase relative to the reference channel n_{ref}, the echo received by the nth channel can be regarded as the self-transmitting and self-receiving echo from the nth equivalent phase center (EPC). The distance between the nth EPC and the reference EPC is $d_n = (n - n_{\text{ref}}) \cdot D/2$, $n = 1, 2, \ldots, N$.

In Fig. 1(a), moving target P is located at $(x_0, y_0, 0)$ with constant radial velocity v_r and azimuth velocity v_a (or named along-track velocity). Fig. 1(b) shows the distance history of the moving target on the slant-range plane. At $\eta = \eta_0$, target P is located at the nearest position A to the satellite, $\Delta\eta$ later, and target P moves to position B. The radial velocity is positive when the target moves away from the satellite. The instantaneous slant range from the moving target to the nth EPC can be described as follows [32, eq. (4)]:

$$\begin{aligned} R_n(\eta) &= \sqrt{[R_0 + v_r(\eta - \eta_0)]^2 + [(V_s - v_a)(\eta - \eta_0) + d_n]^2} \\ &\approx R_0 + v_r(\eta - \eta_0) + \frac{[(V_s - v_a)(\eta + \eta_n - \eta_0)]^2}{2R_0} \end{aligned} \tag{1}$$

where $\eta_n = d_n/(V_s - v_a)$, η is the azimuth time, $R_0 = (H^2 + y_0^2)^{1/2}$ is the shortest slant range. $\eta_0 = x_0/(V_s - v_a)$ is the zero Doppler time, and x_0 is the azimuth position of target. The received echo of the nth EPC is [30, Ch. 5]

$$\begin{aligned} \mathrm{s}_n(\tau, \eta) = A_x \cdot \mathrm{rect}&\left\{\frac{\tau - 2R_n(\eta)/c}{T_r}\right\} w_a(\eta - \eta_0) \\ &\cdot \exp\left\{j\pi K_r\left(\tau - \frac{2R_n(\eta)}{c}\right)^2\right\} \\ &\cdot \exp\left\{-j\frac{4\pi}{\lambda}R_n(\eta)\right\} \end{aligned} \tag{2}$$

where A_x is a complex constant representing the signal gain. τ is the range time, T_r is the pulsewidth, $w_a(\cdot)$ is the azimuth two-way antenna pattern, K_r is the range frequency modulation (FM) rate, c is the velocity of light, and λ is the wavelength. For the sake of narration, we will use "nth channel" to represent the "nth EPC" hereafter.

Range compression is performed on the signal first, and then the nth channel signal is transformed into the range frequency and azimuth time domain, and one can obtain [30]

$$\begin{aligned} \mathrm{S}_n(f_\tau, \eta) = A_x \cdot W_r(f_\tau) w_a(\eta - \eta_0) &\\ \times \exp\{-jkR_0\} \cdot \exp\{-jkv_r(\eta - \eta_0)\} &\\ \times \exp\left\{-jk\frac{[(V_s - v_a)(\eta + \eta_n)]^2}{2R_0}\right\} &\\ \times \exp\left\{-jk\frac{(V_s - v_a)^2[-2(\eta + \eta_n)\eta_0 + \eta_0^2]]}{2R_0}\right\} & \end{aligned} \tag{3}$$

where $k = 4\pi(f_0 + f_\tau)/\lambda$. $W_r(\cdot)$ is the range envelope, f_τ is the range frequency, and f_0 is the carrier frequency.

In the following, we will discuss the unambiguous imaging method of moving target for maritime scenes. Fig. 2 presents

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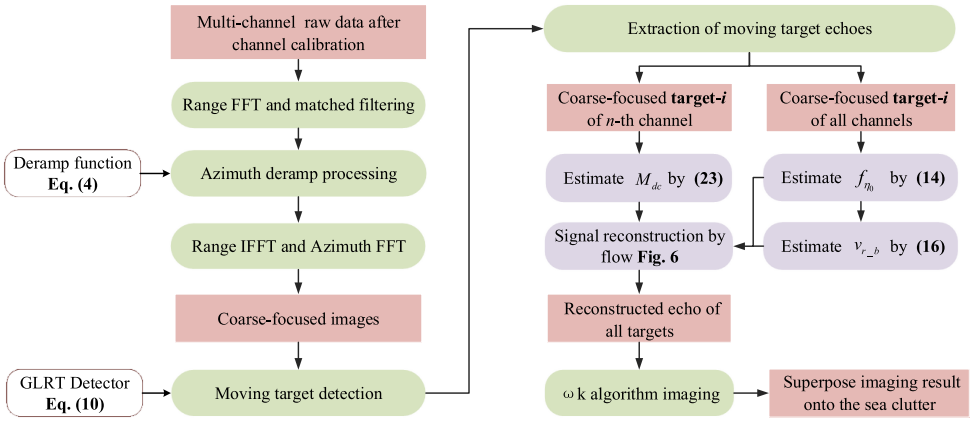


Fig. 2. Data processing flow of the moving target unambiguous imaging method for maritime scenes with the AMC-HRWS SAR system.

the data processing flow, which mainly includes moving target detection, echo extraction, radial velocity estimation, signal reconstruction, and imaging. In Fig. 2, v_{r_b} denotes the baseband radial velocity, i.e., $|v_{r_b}| < \lambda \cdot \mathrm{PRF}/4$, and M_{dc} denotes the ambiguity number of the radial velocity, i.e., the corresponding radial velocity of $M_{\mathrm{dc}} = i$ is $i \cdot \lambda\ \mathrm{PRF}/2$.

III. Radial Velocity Estimation and Signal Reconstruction of Moving Target

Since moving targets have different motion parameters, the echo of the moving target needs to be extracted from the sea clutter and processed separately. In this section, based on azimuth deramp technique, the methods of moving target detection, echo extraction, and radial velocity estimation are presented. Then, we analyze the characteristics of moving target signal in the coarse-focused image domain and propose a novel signal reconstruction method.

A. Azimuth Deramp Processing of Moving Target Signal

According to (3), to eliminate the azimuth phase modulation term, the azimuth deramp function for the nth channel can be constructed as

$$H_n(f_\tau, \eta) = \exp\left\{ jk \frac{[(V_s - v_a)(\eta + \eta_n)]^2}{2R_0} \right\}. \tag{4}$$

After multiplying by (4), the moving target signal of the nth channel can be expressed in the range time and azimuth frequency domain (hereinafter referred to as the deramp frequency domain or coarse-focused image domain) as follows:

$$\begin{aligned} &S_{n,\mathrm{de}}(\tau, f_\eta) \\ &= A_x \cdot \exp\left\{ -j\frac{4\pi}{\lambda} R_{\mathrm{equ}} \right\} \cdot \Phi(\tau, f_\eta) \otimes_2 \\ &\quad \times \left(\sin c\left\{ B_r \left(\tau - \frac{2R_{\mathrm{equ}}}{c} \right) \right\} \cdot \exp\{-j2\pi f_\eta \eta_0\} \right. \\ &\qquad \left. \cdot \mathrm{sinc}\{ T_a (f_\eta + f_{dc} - f_{\eta_0}) \} \cdot \exp\{ j2\pi \cdot f_{\eta_0} \cdot \eta_n \} \right) \end{aligned} \tag{5}$$

where f_η is the azimuth frequency. T_a is the synthetic aperture time, and $f_{\mathrm{dc}} = 2v_r/\lambda$ is the Doppler centroid caused by radial velocity. $R_{\mathrm{equ}} = R_0 - v_r\eta_0 + (V_s - v_a)^2\eta_0^2/2R_0$ is the equivalent slant range. $f_{\eta_0} = 2(V_s - v_a)^2\eta_0/\lambda R_0$ is a single-frequency value after the Doppler spectrum is compressed, which is related to the zero Doppler time of the moving target. $\otimes_2$ represents 2-D convolution, and

$$\begin{aligned} \Phi(\tau, f_\eta) = \int_\eta \int_{f_\tau} &\exp\left\{ j\frac{4\pi f_\tau}{c} \left(v_r - \frac{(V_s - v_a)^2 \eta_0}{R_0} \right) \eta \right\} \\ &\times \exp\{ j2\pi f_\tau \tau - j2\pi f_\eta \eta \} df_\tau d\eta. \end{aligned} \tag{6}$$

Since $v_r \neq 0\ and \eta_0 \neq 0$, phase $\Phi(\tau, f_\eta)$ contains linear RCM, which without compensation will defocus the moving target to a certain extent. However, we can still obtain the coarse-focused moving target in the deramp frequency domain, which is focused on several azimuth frequency and range cells. The coarse-focused signal can be regarded as the 2-D convolution of the ideal-focused signal and the phase $\Phi(\tau, f_\eta)$. According to (5), let function $\mathrm{sinc}(\cdot) = 1$, we can obtain the range and azimuth-focused cell of the moving target, which are $\tau_c = 2R_{\mathrm{equ}}/c$ and $f_{\eta_c}^{(m)} = f_{\eta_0} - f_{\mathrm{dc}}$, respectively.

Due to the low PRF sampling of the HRWS system, the coarse-focused moving target will be folded into the frequency range of $-\mathrm{PRF}/2$ to $\mathrm{PRF}/2$ when the real focused frequency cell satisfies $|f_{\eta_c}^{(m)}| > \mathrm{PRF}/2$, and the azimuth-focused cell becomes $f_{a_c}^{(m)} = f_{\eta_c}^{(m)} - M_1 \cdot \mathrm{PRF}$, $M_1 = \lfloor f_{\eta_c}^{(m)}/\mathrm{PRF} + 0.5 \rfloor$, and $\lfloor \cdot \rfloor$ means round down. In the real case, the absolute value of the radial velocity might be greater than $\lambda\,\mathrm{PRF}/4$, so we divide the radial velocity into the baseband part and the ambiguity part. The ambiguity number is defined as $M_{\mathrm{dc}} = \lfloor f_{\mathrm{dc}}/\mathrm{PRF} + 0.5 \rfloor$, and the baseband radial velocity is $v_{r_b} = v_r - M_{\mathrm{dc}} \cdot v_m$, where $v_m = \lambda\,\mathrm{PRF}/2$. Therefore, the baseband Doppler centroid is $f_{dc_b} = 2\,v_{r_b}/\lambda$, where $|f_{dc_b}| \leq \mathrm{PRF}/2$.

Based on the above analysis, the moving target signal for the nth channel of AMC-HRWS SAR can be rewritten as

$$\begin{aligned} &S_{n,\mathrm{de},l}(\tau, f_a) \\ &= A_x \cdot \exp\left\{ -j\frac{4\pi}{\lambda} R_{\mathrm{equ}} \right\} \cdot \Phi(\tau, f_a) \otimes_2 \\ &\quad \times \left(\sin c\left\{ B_r \left(\tau - \frac{2R_{\mathrm{equ}}}{c} \right) \right\} \cdot \exp\{-j2\pi f_a \eta_0\} \right. \\ &\qquad \left. \cdot \mathrm{sinc}\{ T_a (f_a + l \cdot \mathrm{PRF} - f_{\eta_1}) \} \cdot \exp\{ j2\pi \cdot f_{\eta_0} \cdot \eta_n \} \right) \end{aligned} \tag{7}$$

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where $f_a \in [-\mathrm{PRF}/2, \mathrm{PRF}/2]$. $f_{\eta_1} = f_{\eta_0} - f_{dc_b}$ and $l = \lfloor f_{\eta_1}/\mathrm{PRF} + 0.5 \rfloor$. The azimuth-focused cell of the moving target is $f_{a_c}^{(m)} = f_{\eta_1} - l \cdot \mathrm{PRF}$.

B. Moving Target Detection and Echo Extraction

Azimuth deramp processing is performed on the echoes of each channel to obtain multiple coarse-focused SAR images in the deramp frequency domain. Then, the GLRT algorithm [32], [34] is used to detect moving targets, and their echoes are extracted from each coarse-focused SAR image according to the detection position. First, the samples should be selected to estimate the noise covariance matrix. Since the energy of the moving target is much stronger than the sea clutter, the moving target samples should be removed. A new image $I(a, r)$ is formed by summing the amplitude of all the coarse-focused SAR images, where a and r denote the range and azimuth index, respectively. An amplitude-based threshold is defined as

$$T_{\mathrm{g}} = \mathrm{ave}(I(a,r)) + \sqrt{\sum_{a=1}^{N_a}\sum_{r=1}^{N_r} \frac{[I(a,r) - \mathrm{ave}(I(a,r))]^2}{N_a \cdot N_r - 1}} \quad (8)$$

where ave$(\cdot)$ represents the average amplitude of the image. N_a and N_r are the total number of azimuth and range cell of the image $I(a, r)$, respectively. In the image $I(a, r)$, the positions whose amplitude is lower than the threshold T_g are selected as the sample positions. Then, the noise covariance matrix can be calculated as

$$\mathbf{R}_n = \frac{1}{M}\sum_{m=1}^{M} \mathbf{X}_m \mathbf{X}_m^H \quad (9)$$

where M is the total number of selected samples. $\mathbf{X}_m = [x_1, x_2, \ldots, x_N]^T$ is a complex vector composed of the image values of N coarse-focused SAR images corresponding to the mth sample position. Constructing data vector $\mathbf{P}_{a,r} = [p_1, p_2, \ldots, p_N]^T$, where p_n is the image value of the nth coarse-focused image at position (a, r). The GLRT detector [32] can be constructed as

$$T_{\mathrm{GLRT}}(a,r) = \frac{\mathbf{P}_{a,r}^H \cdot \mathbf{R}_n^{-1} \cdot \mathbf{P}_{a,r}}{1 + \mathbf{P}_{a,r}^H \cdot \mathbf{R}_n^{-1} \cdot \mathbf{P}_{a,r}} \underset{H_0}{\overset{H_1}{\gtrless}} \ell \quad (10)$$

where H_1 assumes that the moving target exists. H_0 means that the moving target does not exist. ℓ is the detection threshold under a certain false alarm, which is a constant for a given sample number and a known statistical distribution.

The moving targets are detected by the GLRT detector, and their echoes can be extracted from each coarse-focused SAR image according to detection positions. After that, the data vector of a moving target can be expressed as follows:

$$\mathbf{S}_{\mathrm{MT}}(\tau, f_a) = \begin{bmatrix} S'_{1,\mathrm{de},l}(\tau, f_a) \\ \vdots \\ S'_{n,\mathrm{de},l}(\tau, f_a) \\ \vdots \\ S'_{N,\mathrm{de},l}(\tau, f_a) \end{bmatrix} + \mathbf{N}_{\mathrm{deramp}}(f_a) \quad (11)$$

where $S'_{n,\mathrm{de},l}(\tau, f_a)$ is the moving target echo of the nth channel in the deramp frequency domain, and $\mathbf{N}_{\mathrm{deramp}}(f_a)$ is the noise vector.

C. Radial Velocity Estimation

1) Estimation of Baseband Radial Velocity v_{r_b}: According to (7), we can obtain the steering vector of the moving target in the deramp frequency domain

$$\mathbf{b}_T = \begin{bmatrix} \exp\{j2\pi \cdot f_{\eta_0} \cdot \eta_1\} \\ \vdots \\ \exp\{j2\pi \cdot f_{\eta_0} \cdot \eta_n\} \\ \vdots \\ \exp\{j2\pi \cdot f_{\eta_0} \cdot \eta_N\} \end{bmatrix}. \quad (12)$$

According to the analysis in Part-A, f_{η_0} is a constant related to the azimuth-focused cell $f_{a_c}^{(m)}$ and baseband radial velocity v_{r_b} of the moving target. The value of $f_{a_c}^{(m)}$ can be known from the detection result; as long as the value of f_{η_0} is estimated, v_{r_b} can be obtained. Based on Capon's [35] spatial spectrum estimation algorithm, the estimation problem of f_{η_0} can be described as follows:

$$\begin{cases} \min E\{P(f_{\eta_0})\} \Leftrightarrow \min \mathbf{W}^H \mathbf{R}_T(f_a) \mathbf{W} \\ \text{s.t} \quad \left|\mathbf{W}^H \mathbf{b}_T\right|^2 = 1. \end{cases} \quad (13)$$

The output power is

$$P(f_{\eta_0}) = \frac{1}{\mathbf{b}_T^H \mathbf{R}_T^{-1}(f_a) \mathbf{b}_T}. \quad (14)$$

$\mathbf{R}_T(f_a)$ is the covariance matrix of the moving target data

$$\begin{aligned} \mathbf{R}_T(f_a) &= E\{\mathbf{S}_{\mathbf{MT}}(\tau, f_a) \mathbf{S}_{\mathbf{MT}}(\tau, f_a)^H\} \\ &= \frac{1}{J}\sum_{l_a}\sum_{l_r} \mathbf{S}_{\mathbf{MT}}(\tau^{l_r}, f_a^{l_a}) \mathbf{S}_{\mathbf{MT}}(\tau^{l_r}, f_a^{l_a})^H \end{aligned} \quad (15)$$

where J is the total number of data cells in the area where the moving target is located, which is determined by the GLRT detection result. l_a and l_r are the index of the azimuth and range cell, respectively.

The estimated value $\hat{f_{\eta_0}}$ can be obtained by maximizing the output power. It can be known from (7) that $f_{a_c}^{(m)} = f_{\eta_0} - f_{dc_b} - l \cdot \mathrm{PRF}$, so v_{r_b} can be calculated as follows:

$$v_{r_\mathrm{b}} = \frac{\left(\hat{f_{\eta_0}} - f_{a_c}^{(m)} - l \cdot \mathrm{PRF}\right) \cdot \lambda}{2} \quad (16)$$

where $l = \lfloor (\hat{f_{\eta_0}} - f_{a_c}^{(m)}/\mathrm{PRF}) + (1/2) \rfloor$.

2) Estimation of Ambiguity Number M_{dc}: In this part, we will present the method of M_{dc} estimation. Taking the extracted moving target echo of the nth channel, i.e., $S'_{n,\mathrm{de},l}(\tau, f_a)$, the azimuth inverse fast Fourier transform (IFFT), range fast Fourier transform (FFT), and inverse azimuth deramp processing are carried out on it. The corresponding inverse azimuth deramp function is

$$H_{\mathrm{inv_}n}(f_\tau, \eta) = \exp\left\{-jk\frac{[(V_s - v_a)(\eta + \eta_n)]^2}{2R_0}\right\}. \quad (17)$$

After multiplying by (17), the moving target signal of the nth channel is

$$\begin{aligned} S_{\mathrm{MT},n}(f_\tau, \eta) = {} & A_x \cdot W_r(f_\tau) w_a(\eta - \eta_0) \\ & \cdot \exp\{-jkR_0\} \cdot \exp\{-jkv_r(\eta - \eta_0)\} \\ & \cdot \exp\left\{-jk\frac{[(V_s - v_a)(\eta + \eta_n - \eta_0)]^2}{2R_0}\right\}. \end{aligned} \quad (18)$$

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Second, the zero Doppler time η_0 is calculated by $\hat{f}_{\eta_0}$. Then azimuth deramp processing is performed on signal (18) with a new reference function

$$H_{\mathrm{MT}}(f_\tau, \eta) = \exp\left\{ jk\frac{[(V_s - v_a)(\eta + \eta_n - \eta_0)]^2}{2R_0} \right\}. \quad (19)$$

The second exponential term of (18) contains the linear RCM caused by the radial velocity, which will defocus the moving target if without compensation. In the case that v_{r_b} has been estimated, we construct a compensating phase function

$$H_{v_r}(f_\tau, \eta) = \exp\left\{ jk\left(M_a v_m + v_{r_\mathrm{b}}\right)(\eta - \eta_0)\right\} \quad (20)$$

where M_a is the assumed value of M_{dc}. After multiplying by phase (19) and (20), range IFFT and azimuth FFT are performed on the signal. Then, the moving target signal in the deramp frequency domain is obtained

$$\begin{aligned} &S_{M_a}(\tau, f_a) \\ &= A_x \cdot \left(\sin c\left\{ B_r\left(\tau - \frac{2R_0}{c} \right) \right\} \cdot \sin c\{T_a f_a\} \right) \\ &\quad \times \exp\left\{ -j\frac{4\pi}{\lambda} R_0 \right\} \exp\{-j2\pi f_a \eta_0\} \otimes_2 \Phi_{M_a}(\tau, f_a). \end{aligned} \quad (21)$$

Now, the azimuth-focused cell of the moving target is $f_{a_c}^{(m)} = 0$, where

$$\begin{aligned} \Phi_{M_a}(\tau, f_a) = \int_\eta \int_{f_\tau} &\exp\{jk(M_a - M_{\mathrm{dc}}) \cdot v_m(\eta - \eta_0)\} \\ &\times \exp\left\{ j2\pi f_\tau \tau - j2\pi f_\eta \eta \right\} df_\eta d\eta. \end{aligned} \quad (22)$$

The signal $S_{M_a}(\tau, f_a)$ can be regarded as the 2-D convolution of the ideal-focused signal and the phase $\Phi_{M_a}(\tau, f_a)$. If $M_a = M_{\mathrm{dc}}$, the best focused target can be obtained. The entropy value [36] can be used to evaluate the quality of the focused result, which is defined as

$$E\left(S_{M_a}\right) = \int_\tau \int_{f_a} \frac{\left|S_{M_a}(\tau, f_a)\right|^2}{S_e} \log \frac{\left|S_{M_a}(\tau, f_a)\right|^2}{S_e} df_a d\tau \quad (23)$$

where $S_e = \int_\tau \int_{f_a} |S_{M_a}(\tau, f_a)|^2 df_a d\tau$. The small entropy value means high quality of the focused result. Hence, the ambiguity number M_{dc} can be obtained by minimizing $E(S_{M_a})$.

D. Signal Reconstruction and Imaging

The signal model of the moving target is related to motion parameters, so that the moving target in the observation scene cannot be reconstructed correctly together with the stationary scene. To reconstruct the moving target signal correctly, the motion parameters must be considered. For spaceborne SAR, $v_a \ll V_s$, the influence of azimuth velocity on signal reconstruction is analyzed in [24], and the experimental results are given, which proves that v_a can be ignored. Radial velocity is the key parameter of moving target signal reconstruction.

Without considering the azimuth velocity, the difference between the moving target signal and ST signal is mainly reflected in the phase term related to the radial velocity, i.e., the second exponential term of (18)

$$\begin{aligned} H_{v_r}(f_\tau, \eta) = &\exp\left\{ -j2\pi \cdot \frac{2v_r}{\lambda} \cdot (\eta - \eta_0) \right\} \\ &\times \exp\left\{ -j\frac{4\pi f_\tau}{c} v_r(\eta - \eta_0) \right\}. \end{aligned} \quad (24)$$

The first exponential term of (24) is the linear phase related to azimuth time, which makes the Doppler spectrum center of the signal deviate from the zero frequency, resulting in the failure of signal reconstruction and the wrong focused position of the real target. The second exponential term of (24) is the linear RCM, resulting in the defocus of the moving target. To reconstruct the moving target signal correctly, there are two common methods.

1) Phase (24) is compensated first, and then the signal is reconstructed by the ST reconstruction methods, such as Krieger's reconstruction filtering method [13] and STAP-based method [15].
2) The radial velocity is directly introduced into the reconstruction filter to reconstruct the signal [21].

In essence, the above methods use the space–time spectrum relationship of the signal to reconstruct the unambiguous Doppler spectrum of the signal, which requires the echo data of multiple channels. There are two problems in the reconstruction of moving target signal: 1) the correct space–time spectrum relationship cannot be obtained when the estimated radial velocity is not accurate, which will degrade the reconstruction performance and generate multiple false targets and 2) for real data, it is difficult to completely eliminate the phase error between channels (channel imbalance), and the residual phase error (RPE) also lead to false targets [37].

Azimuth ambiguity-to-signal ratio (AASR) is defined to evaluate the imaging quality of the reconstructed signal

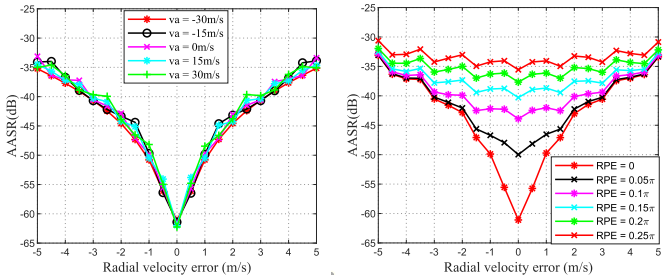
$$\mathrm{AASR} = 10 \cdot \log 10 \frac{P_{\mathrm{amb}}}{P_m} \quad (25)$$

where P_{amb} is the maximum power of false targets, and P_m is the power of the real target. According to the spaceborne parameters in Table I, the point target simulation experiment is conducted. Without considering the RPE between channels, Fig. 3(a) shows the relationship between AASR and radial velocity error under different azimuth velocities. If the radial velocity error is greater than 3 m/s, AASR will be greater

TABLE I
SAR SYSTEM PARAMETERS

Symbol	Parameters	Value
V_s	Platform velocity	7508 m/s
f_0	Carrier frequency	9.6 GHz
T_r	Pluse duration	50 μs
B_r	Signal bandwidth	100 MHz
L_a	Antenna length	10 m
N	Number of channels	5
Ba	Doppler bandwidth	6652.1 Hz
PRF	Pulse repetition frequency	1550 Hz
R_c	Slant-range of scene center	823.25 km
η_{max}	$\mathrm{PRF_e}$ time	1.7992 s

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(a) (b)

Fig. 3. (a) Influence of radial velocity error and azimuth velocity on AASR. (b) Influence of radial velocity error and RPEs on AASR.

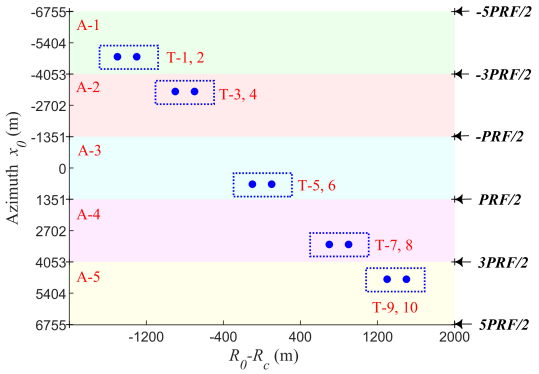
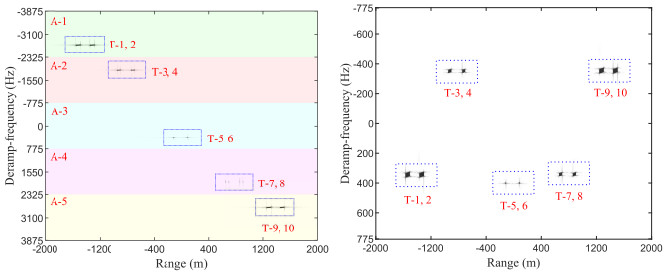


Fig. 4. Imaging scene.



(a) (b)

Fig. 5. Coarse-focused image of STs in the imaging scene. (a) Azimuth sampling rate is PRF_e. (b) Azimuth sampling rate is PRF.

than -40 dB, and false targets may affect the image quality. The influence of azimuth velocity on signal reconstruction is very small. Actually, although azimuth velocity does not affect the signal reconstruction, it causes the defocus of the real target, reducing the maximum power of the real target. When there are false targets, the value of AASR will increase. Next, the RPE of the reference channel is set to 0 radian, and the RPE of other channels is set to α radian. Fig. 3(b) shows the relationship between AASR and radial velocity error under different RPEs; as can be seen that when the RPE is greater than $0.15\,\pi$ radian, AASR will be greater than -40 dB even if there is no error in the radial velocity.

To solve these problems, we propose a moving target signal reconstruction method with better performance based on the characteristics of moving target signal in the coarse-focused image domain. Similarly, the phase (24) causes the focused position of the moving target in the coarse-focused image domain to shift f_{dc}, i.e., $f_{\eta_c}^{(m)} = f_{\eta_0} - f_{\mathrm{dc}}$, and also leads to the defocus of the moving target. Therefore, phase (24) should be compensated before reconstructing the moving target signal. For clearer description, ST is taken as an example to discuss the signal reconstruction method, and then the step of compensating phase (24) is added into the processing flow to achieve the signal reconstruction of moving target.

For ST, $f_{\mathrm{dc}} = 0$, its focused position in the deramp frequency domain is $f_{\eta_c}^{(s)} = f_{\eta_0} = 2{V_s}^2\eta_0/\lambda R_0$. The PRF of the HRWS system is lower than the azimuth signal bandwidth B_a. For coarse-focused targets with $|f_{\eta_c}^{(s)}| > \mathrm{PRF}/2$, they will be folded into the frequency range of $-\mathrm{PRF}/2$ to $\mathrm{PRF}/2$ with a period of PRF in the deramp frequency domain. Its azimuth-focused cell becomes $f_{a_c}^{(s)} = f_{\eta_0} - M \cdot \mathrm{PRF}$, $M = \lfloor f_{\eta_0}/\mathrm{PRF} \rfloor$. Assuming that the Doppler ambiguity number is $M_{\mathrm{hrws}} = \lceil B_a/\mathrm{PRF} \rceil = 5$, the equivalent PRF_e of the reconstructed signal is $5 \cdot \mathrm{PRF}$.

Taking the system parameter in Table I as an example, an imaging scene with azimuth time range of $-5\eta_{\max}/2$ to $5\eta_{\max}/2$ (where $\eta_{\max} = \mathrm{PRF}_e \cdot \lambda R_0/2{V_s}^2$) is set, which is shown in Fig. 4. The whole scene is divided into five areas from A-1 to A-5, and the azimuth width of each area is $V_s \cdot \eta_{\max}/5$ m, corresponding to a deramp frequency interval with the size of PRF. Two STs are placed in each area. If the azimuth sampling rate of the system is PRF_e, after performing range compression and azimuth deramp processing on the echo of imaging scene, the coarse-focused result of targets can be obtained, as shown in Fig. 5(a). Comparing Fig. 4 and Fig. 5(a), it can be seen that the focused position of the target in the deramp frequency domain corresponds to its zero Doppler time one to one. If the azimuth sampling rate is PRF, the coarse-focused result of areas A-1, A-2, A-4, and A-5 will be folded into area A-3, as shown in Fig. 5(b).

Comparing Fig. 5(a) and (b), the following conclusions can be obtained after analysis: In the case that only aliased coarse-focused image can be obtained, as long as the parameter f_{η_0} of each coarse-focused target can be estimated, it can determine which area (or frequency interval) of the image (Fig. 5(a)) these targets are originally focused on, so as to restore the targets (that are) folded in A-3 to the correct position, i.e., restore Fig. 5(b) to Fig. 5(a). After that, inverse azimuth deramp operation is performed on unfolded coarse-focused image to obtain the unfolded echo. STs in real imaging scene are continuously distributed, so it is impossible to estimate the parameter f_{η_0} of each ST. However, moving targets are sparsely distributed, and their parameter f_{η_0} can be estimated by the method proposed in Section III-C. It is possible to unfold the folded coarse-focused target. The specific process is as follows.

1) Take the coarse-focused image $S_{n,\mathrm{de},l}(\tau, f_a)$ of the nth channel, i.e., (7), and assume that the image size is $N_a \times N_r$ (azimuth $\times$ range).
2) According to the GLRT detection result, the complex image value at the location of target T-i is retained, and

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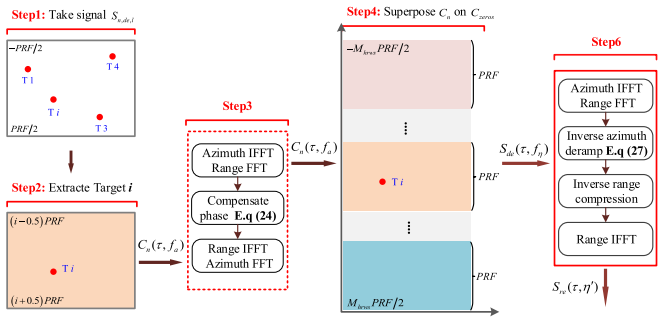


Fig. 6. Flowchart of moving target signal reconstruction.

the other positions are set to zero. Thus, a new complex image $C_n(\tau, f_a)$ is obtained, and the size is also $N_a \times N_r$.

3) The phase term (24) related to the radial velocity of the moving target is compensated in the range frequency and azimuth time domain

$$S(f_\tau, \eta) = \mathrm{FFT}_{\mathrm{rg}}\{\mathrm{IFFT}_{\mathrm{az}}\{C_n(\tau, f_a)\}\} \cdot H_{\mathrm{com}}$$
$$C_n(\tau, f_a) = \mathrm{FFT}_{\mathrm{az}}\left\{\mathrm{IFFT}_{\mathrm{rg}}\{S(f_\tau, \eta)\}\right\} \quad (26)$$

where $\mathrm{FFT}_{\mathrm{az}}$ and $\mathrm{FFT}_{\mathrm{rg}}$ represent the range and azimuth Fourier transforms, respectively. The phase function H_{com} is the conjugate of $H_{v_r}(f_\tau, \eta)$.

4) A zero-valued matrix $C_{\mathrm{zeros}}(\tau, f_\eta)$ is generated, with size of $(M_{\mathrm{hrws}} \cdot N_a) \times N_r$. The matrix is evenly divided into M_{hrws} submatrices along the azimuth, corresponding to M_{hrws} deramp frequency intervals: $[i - 1/2, i + 1/2] \cdot \mathrm{PRF}$, $i = (1 - M_{\mathrm{hrws}})/2, \ldots, (M_{\mathrm{hrws}} - 1)/2$. According to the estimated value $\hat{f}_{\eta_0}$ of the target, confirm the frequency interval where $\hat{f}_{\eta_0}$ is located and superpose $C_n(\tau, f_a)$ onto the matching area in $C_{\mathrm{zeros}}(\tau, f_\eta)$.

5) Repeat steps 1)–4) until all the targets are processed, and the unfolded coarse-focused image $S_{\mathrm{de}}(\tau, f_\eta)$ can be obtained. If M_{hrws} is even, in step 3), a zero-valued matrix with size of $[(M_{\mathrm{hrws}} + 1) \cdot N_a] \times N_r$ is generated, which is evenly divided into $M_{\mathrm{hrws}} + 1$ submatrices along the azimuth. The data with size $N_a/2 \times N_r$ at both azimuth ends of $S_{\mathrm{de}}(\tau, f_\eta)$ should be discarded.

6) Azimuth IFFT and range FFT are performed on the signal $S_{\mathrm{de}}(\tau, f_\eta)$, and then inverse azimuth deramp processing is performed. Finally, inverse range compression and range IFFT are performed to obtain the reconstructed echo $S_{re}(\tau, \eta')$. The inverse azimuth deramp function is

$$H_{\mathrm{inv,de}}(f_\tau, \eta') = \exp\left\{-jk\frac{\left[(V_s - v_a)(\eta' + \eta_n)\right]^2}{2R_0}\right\} \quad (27)$$

where η' represents the new azimuth time axis, and the time sampling interval is $1/M_{\mathrm{hrws}}$ of the original.

Fig. 6 takes the target T-i as an example and illustrates the processing flow. In the deramp frequency domain, the Doppler spectrum of the signal has been compressed. Zero filling in the deramp frequency domain is done to increase the frequency range of the signal, and then inverse azimuth deramp operation is performed on the signal using function (27).

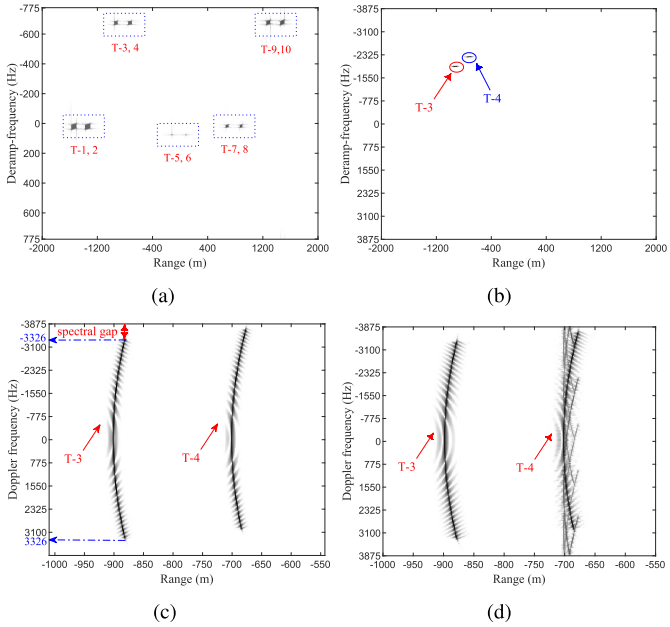


Fig. 7. (a) Aliased coarse-focused image of moving targets. ($v_r = 5$ m/s). (b) Coarse-focused reconstructed signal of T-3 and T-4 obtained by the proposed method. The reconstructed signal of T-3 and T-4 in the R-D domain obtained (c) by the proposed method and (d) by the reconstruction filter method.

This is equivalent to using a new function whose time sampling interval is $1/M_{\mathrm{hrws}}$ times of the original to carry out azimuth FM of the signal, thus improving the time sampling rate and realizing signal reconstruction. The equivalent PRF_e of the reconstructed signal is $M_{\mathrm{hrws}} \cdot \mathrm{PRF}$, and the oversampling rate of the azimuth signal is $r_{\mathrm{az}} = \mathrm{PRF}_e/B_a$. If r_{az} is large, the reconstructed signal has a wide Doppler spectrum gap, which means that the Doppler spectrum center of the signal can be allowed to move within a certain range without causing spectrum rupture, so there will not be false targets in the imaging result. In other words, if there is an error in the estimated radial velocity that makes the inaccurate compensation for phase (24), it will lead to the defocus of the moving target and the offset of the azimuth-focused position, but cannot cause false targets. The allowable radial velocity error is

$$\mathrm{Av}_r = \left|\frac{(\mathrm{PRF}_e - B_a)}{2} \cdot \frac{\lambda}{2}\right|. \quad (28)$$

Since the focused position $f_{a_c}^{(m)}$ of the moving target in the coarse-focused image is determined by v_{r_b} and f_{η_0}, allowing an error in the radial velocity is equivalent to allowing an error in the estimation result of f_{η_0}. The presence of a δf_{η_0} error in f_{η_0} is equivalent to the uncompensated residual phase $\exp\{-j2\pi \cdot \delta f_{\eta_0} \cdot (\eta - \eta_0)\}$ in the first exponential term of phase (24). The radial velocity of all the targets in Fig. 4 is set to 5 m/s, and the coarse-focused image is shown in Fig. 7(a). Compared with Fig. 5(b), it can be seen that the focused positions of the moving targets shift to the negative frequency axis, and the defocus of the target is more serious. Only taking T-3 and T4 in Fig. 7(a) as an example, and processing them with the flow of Fig. 6, the result $S_{\mathrm{de}}(\tau, f_\eta)$ is as shown in

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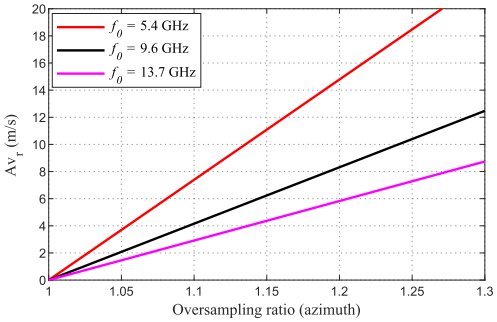


Fig. 8. Relationship of Av$_r$ and r_{az} under three typical carrier frequencies.

TABLE II
SYSTEM PARAMETERS FOR THE SIMULATED DATA

Symbol	Parameters	Value
V_s	Platform velocity	7508 m/s
f_0	Carrier frequency	9.6 GHz
T_r	Pluse duration	50 μs
B_r	Range bandwidth	100 MHz
L_a	Antenna length	10 m
N	Number of channels	4
Ba	Doppler bandwidth	5321.7 Hz
PRF	Pulse repetition frequency	1526.6 Hz
φ_{min}	Minimum look angle	28.25°
φ_{max}	Maximum look angle	34.53°
R_c	Slant-range of scene center	837.53 km
v_{rmax}	Maximum v_{r_b}	11.93 m/s

Fig. 7(b). Among them, T-3 accurately compensates phase (24), while T-4 does not, i.e., the radial velocity error is 5 m/s. Therefore, the focused position of T4 is offset relative to T-3. Reverse azimuth deramp operation is performed on the signal, and then transformed into the R-D domain. The result is shown in Fig. 7(c). It can be seen that the Doppler spectrum of two targets is reconstructed correctly, and there are obvious spectral gaps. The Doppler center of T-3 is zero, and the spectral gap width at both ends is the same. The Doppler center of T-4 deviates from zero frequency, but the spectrum is still continuous. For comparison, Fig. 7(d) shows the result processed by the reconstruction filtering method, in which the Doppler spectrum of T-4 is not reconstructed correctly. According to the parameters in Table I, the relationship between Av$_r$ and r_{az} under three typical carrier frequencies is given in Fig. 8. The greater the PRF of the system, the greater the Av$_r$.

The proposed signal reconstruction method can eliminate false targets even if there is an error in the radial velocity when the focusing effect and positioning of the target are not considered. In addition, since only the data of single channel are required, false targets caused by unbalanced channels can also be avoided. In the worst case that the radial velocity error is large or the spectrum gap is narrow, thus resulting in spectrum rupture, it will only lead to one false target. The ωk-algorithm is used for imaging, so as to obtain the unambiguous image of moving targets. Since the purpose of our article is to solve the unambiguous imaging problem of the moving target for HRWS SAR, the precise focusing problem of the moving target is not discussed, so only the estimation of radial velocity is considered. Precision focusing can be achieved with autofocus technology or other technologies.

IV. DISCUSSION OF AZIMUTH DERAMP PROCESSING

A. Moving Target Focused on the Azimuth Edge of the Coarse-Focused SAR Image

The coarse-focused moving target will be split at the azimuth edge of the coarse-focused image if $f_{a_c}^{(m)}$ is close to $\pm$PRF/2. It is not conducive to moving target extraction. Although the probability of this situation is very low, it is necessary to analyze.

This problem can be solved by multiplying a linear phase in the azimuth time domain, which is equivalent to a frequency shift. The moving target can be moved near the azimuth center of the coarse-focused image. This linear phase is

$$H_{\text{offset}}(\eta) = \exp\left\{ j2\pi \frac{\text{PRF}}{2}\eta \right\}. \tag{29}$$

The frequency shift has no influence on the estimation of f_{η_0}, but should be considered when estimating v_{r_b}

$$v_{r_\text{b}} = \frac{\left(\hat{f_{\eta_0}} + \text{PRF}/2 - f'_{a_c} - l \cdot \text{PRF}\right) \cdot \lambda}{2} \tag{30}$$

where $l = \lfloor (\hat{f_{\eta_0}} - f'_{a_c})/\text{PRF} \rfloor + 1$, and f'_{a_c} is the new azimuth-focused cell. Before estimating M_{dc}, the linear phase (29) should be compensated. In step (4) of the signal reconstruction flow, the frequency interval where $\hat{f_{\eta_0}} + \text{PRF}/2$ is located shall be determined, and superposing the image $C_n(\tau, f_a)$ onto the matching area in $C_{\text{zeros}}(\tau, f_\eta)$. Then, the phase (29) should be compensated for signal $C_{\text{zeros}}(\tau, f_\eta)$.

B. Difficulty of Moving Target Detection in the Coarse-Focused Image

According to (4), azimuth deramp function takes $\eta = 0$ as a reference, which results in residual linear RCM for the targets far away from the azimuth scene center. For the moving target, the radial velocity also results in linear RCM. To review Section III-A, the linear RCM is contained in phase $\Phi(\tau, f_\eta)$, i.e., formula (6). Solving the integral of $\Phi(\tau, f_\eta)$ with respect to f_τ, we can obtain

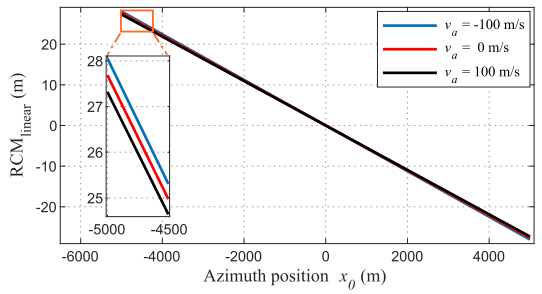
$$\Phi(\tau, f_\eta) = \int_\eta \text{sinc}\left\{ B_r \left(\tau - \frac{2}{c}\Delta R \right) \right\} \exp\{-j2\pi f_\eta \eta\} d\eta \tag{31}$$

where $\Delta R = [v_r - (V_s - v_a)x_0/R_0]\eta$, $x_0 = \eta_0 \cdot (V_s - v_a)$ is the azimuth position of the target. ΔR varies with the slow time, and the maximum linear RCM can be calculated as follows:

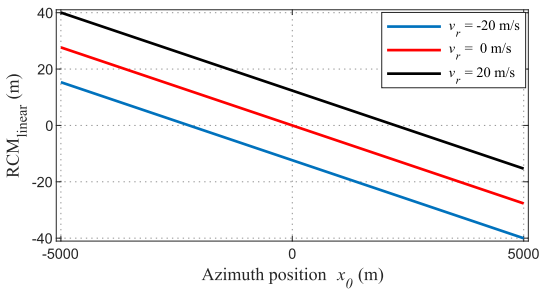
$$\text{RCM}_{\text{linear}} = \Delta R|_{\eta = T_a/2}. \tag{32}$$

The RCM$_{\text{linear}}$ is related to v_r, v_a, and x_0. For certain moving target, its motion parameters are immutable unknowns. Based on the spaceborne parameters in Table II, we first assume that $v_r = 0\ m/s$, and the relationship graph of RCM$_{\text{linear}}$ and x_0 under different v_a is plotted in Fig. 9(a). Then assuming that $v_a = 0$ m/s, the relationship graph of RCM$_{\text{linear}}$ and x_0 under different v_r is plotted in Fig. 9(b). The results show

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(a)



(b)

Fig. 9. Relationship of $\text{RCM}_{\text{linear}}$ and x_0. (a) Under different v_a ($v_r = 0$ m/s). (b) Under different v_r ($v_a = 0$ m/s).

that: 1) linear RCM is mainly caused by v_r and x_0 and 2) the farther the moving target is from the azimuth center, the larger the $\text{RCM}_{\text{linear}}$ is, so that the more serious the defocus is. Therefore, to facilitate moving target detection, the scene size needs to be limited.

In addition, there is a scene folding problem for the coarse-focused image. It is assumed that two moving targets are located at the same slant range gate, if the difference between their real azimuth-focused cells $f_{\eta_c}^{(m)}$ is equal to an integer multiple of PRF, and the two targets will overlap in the coarse-focused image so that their echoes cannot be separated. The moving targets with different shortest slant range can be distinguished in the image. Provided that there are many moving targets within the same slant range gate, it would increase the probability of moving targets folding together, which is not conducive to echo extraction. Based on the above analysis, the echo data should be partitioned into several segments along the azimuth according to the equivalent PRF time, i.e., PRF_{e} time.

C. Mismatch of Azimuth Deramp Function

For moving targets, their shortest slant range R_0 and azimuth velocity v_a are unknowns, which causes two problems: 1) (4) is difficult to construct accurately and 2) parameter η_n used in steering vector $\mathbf{b}_T$ is not accurate. Specifically, the mismatch problem is analyzed as follows.

In actual data processing, R_0 takes the shortest slant range of the scene center R_c and $v_a = 0$; the actual azimuth deramp function is

$$H_{n_\text{real}}(f_\tau, \eta) = \exp\left\{ j\ k\frac{\alpha(V_s - v_a)^2}{2R_0}(\eta + \eta_n)^2 \right\} \tag{33}$$

where $\alpha = (R_0/R_c) \cdot (V_s^2/(V_s - v_a)^2)$ is called the mismatch factor.

Multiplying signal (3) with function (33), and transforming it into deramp frequency domain, one can obtain

$$S_{n,\text{de_mis}}(\tau, f_\eta) = S_{n,\text{de}}(\tau, f_\eta) \otimes_2 \Phi_{n,mis}(\tau, f_\eta) \tag{34}$$

where

$$\begin{aligned}\Phi_{n,\text{mis}}(\tau, f_\eta) &= \int_\eta \int_{f_\tau} \exp\left\{ -j\frac{4(f_0 + f_\tau)}{c}\frac{(1-\alpha)(V_s - v_a)^2}{2R_0}(\eta + \eta_n)^2 \right\} \\ &\quad \times \exp\{ j2\pi f_\tau \tau - j2\pi f_\eta \eta \} d f_\tau d\eta. \end{aligned} \tag{35}$$

Solving the integral of $\Phi_{n,mis}(\tau, f_\eta)$ with respect to f_τ, we can obtain

$$\begin{aligned}\Phi_{n,\text{mis}}(\tau, f_\eta) = \int_\eta \text{sinc}\left\{ B_r\left(\tau - \frac{2}{c} \cdot \Delta R_{n,\text{mis}} \right) \right\} \\ \times \psi_n(\eta) \cdot \exp\{ -j2\pi f_\eta \eta \} d\eta \end{aligned} \tag{36}$$

where

$$\psi_n(\eta) = \exp\{ -j\pi K_a'(\eta^2 + 2\eta_n \eta + \eta_n^2) \} \tag{37}$$

and

$$\Delta R_{n,\text{mis}} = \frac{\lambda}{4} K_a' (\eta + \eta_n)^2 \tag{38}$$

and

$$K_a' = \frac{2(1-\alpha)(V_s - v_a)^2}{\lambda R_0} = \frac{2(\text{V}_\text{s} - \text{v}_\text{a})^2}{\lambda R_0} - \frac{2\text{V}_\text{s}^2}{\lambda R_c}. \tag{39}$$

In phase $\psi_n(\eta)$, the η^2 term is the azimuth modulation term with a minor frequency FM K_a'. The η term is a linear phase associated with the channel, which corresponds to a frequency shift $\partial f_n = K_a' \cdot \eta_n$ in the frequency domain. The η_n^2 term can be rewritten as $\exp\{ j2\pi \cdot (-\partial f_n/2) \cdot \eta_n \}$. It can be regarded as introducing an error $\partial f_n/2$ into the estimation result of f_{η_0}.

According to the parameters in Table II, and assuming that $v_a \in [\ -40, 40\ ]$ m/s and $R_0 \in R_c + [\ -3T_r/2,\ 3T_r/2\ ] \cdot c/2$. The relationship between K_a', R_0, and v_a is shown in Fig. 10. When v_a and R_0 take the minimum or maximum value simultaneously, the corresponding K_a' value is the maximum. In the following calculation, the maximum value of K_a', 105.29 Hz/s, is taken as an example. For the first channel, $\partial f_1 = -0.026$ Hz. For the Nth channel, $\partial f_N = 0.026$ Hz. The frequency sampling interval is

$$\partial f = \frac{PRF}{N_a} = \frac{PRF}{PRF \cdot T_s} = \frac{1}{T_s} \tag{40}$$

where N_a is the number of azimuth samples. T_s is the azimuth time for collecting the echo of the selected scene.

Taking the echo of a PRF_e time, the value of ∂f is 0.705 Hz. Since $\partial f_N - \partial f_1 \ll \partial f$ and $\partial f_N \ll \partial f$, η and η_n^2 terms can be ignored. Phase $\Phi_{n,mis}(\tau, f_\eta)$ can be considered as the Fourier transform result of a minor FM rate signal. According to the minor FM rate signal theorem [38], $\Phi_{n,mis}(\tau, f_\eta)$ can be approximately regarded as a symmetrical azimuth defocus signal. As can be seen from Fig. 10, the target is further away from the scene center of the slant range plane, and K_a' is larger and the azimuth defocus is more serious. Therefore, the echo

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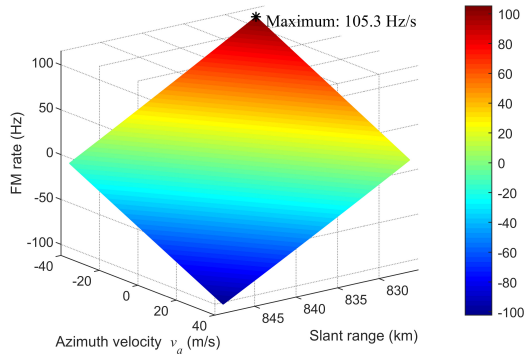


Fig. 10. Relationship between FM rate K_a', slant range R_0, and azimuth velocity v_a.

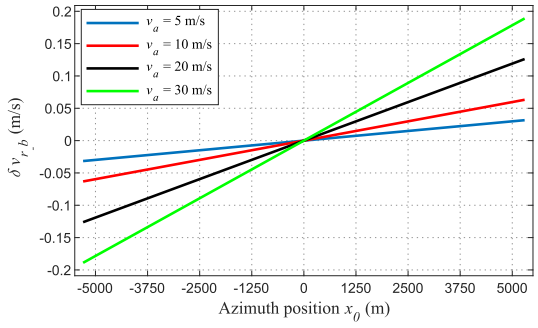


Fig. 11. Relationship between error δv_{r_b} and parameter x_0 under different azimuth velocities.

data can also be divided into several segments along the range according to the pulse width.

$\Delta R_{n,\mathrm{mis}}$ varies with η and contains residual quadratic RCM, and the maximum residual RCM is

$$\mathrm{RCM}_{n,\mathrm{res}} = \Delta R_{n,\mathrm{mis}} \mid_{\eta=T_a/2}. \tag{41}$$

The $\mathrm{RCM}_{n,\mathrm{res}}$ of the first channel and the Nth channel are 0.5079 and 0.5083 m, respectively. For high-resolution SAR systems, the residual RCM will defocus the targets. However, the difference in $\mathrm{RCM}_{n,\mathrm{res}}$ between the Nth channel and the first channel is 4×10^{-4} m, which is much smaller than the sampling cell, so the difference between channels can be ignored. Based on the above analysis, the mismatch result can be regarded as 2-D convolution of the ideal azimuth deramp result $S_{n,\mathrm{de}}(\tau, f_\eta)$ and the 2-D defocused signal $\Phi_{n,\mathrm{mis}}(\tau, f_\eta)$, and the influence on the estimation of v_{r_b} and M_{dc} can be ignored.

Since the azimuth velocity is unknown, the parameter η_n in (33) and steering vector $\mathbf{b}_T$ are actually taken as d_n/V_s, introducing another mismatch factor: $\beta = (V_s - v_a)/V_s$. According to the previous analysis, the phase terms related to η_n in phase $\Phi_{n,\mathrm{mis}}(\tau, f_\eta)$ can be ignored, and thus introducing factor β with a value close to "1" into parameter η_n does not affect the analysis results. β mainly causes an error in the estimation of f_{η_0}

$$\delta f_{\eta_0} = [1-\beta] f_{\eta_0} \approx \frac{2x_0}{\lambda R_0} v_a. \tag{42}$$

The error of v_{r_b} is $\delta v_{r_b} = \lambda \cdot \delta f_{\eta_0}/2$. Based on the parameters in Table II, the relationship graph of δv_{r_b} and x_0 under different v_a is plotted in Fig. 11, where the range of x_0 corresponds to a PRF_e time, and the range of v_a is ± 30 m/s. The maximum error of the radial velocity is near 0.2 m/s. In most cases, the speed range of the ship is 12–35 knots, i.e., 6–18 m/s, and since the azimuth velocity is the azimuth component of the ship's speed over ground (SOG), the actual azimuth velocity is smaller. The radial velocity error caused by unknown azimuth velocity can be controlled within 0.1 m/s in most cases.

V. Experimental Results

In this section, we first conduct an experiment to evaluate the performance of the radial velocity estimation method under different distribution models of sea clutter. Then, the methods of moving target detection, echo extraction, radial velocity estimation, and signal reconstruction are verified by the four-channel simulated data. Finally, the real data of GaoFen-3 SAR with ultrafine stripmap (UFS) mode are processed by the proposed method.

A. Performance Evaluation of the Radial Velocity Estimation Method

Moving target detection for marine scenes is one of the important applications of AMC-HRWS SAR. The influence of sea clutter on radial velocity estimation should be considered. In the simulation experiment, four kinds of sea clutter models are simulated as the background of the detection scene. The amplitude probability density function of the homogeneous scene is usually approximated by Rayleigh distribution. Weibull and Gamma distributions are commonly used in the heterogeneous scene. For scenes with high spatial resolution, the heavy-tailed K-distribution [39] is widely used. The system parameters in Table II are used in this section. The PRF of the system is set to nonuniform sampling, and the Doppler ambiguity number is $M_{\mathrm{hrws}} = 4$.

First, the effectiveness of the proposed radial velocity estimation method is verified. We simulate the echo of a moving target whose $x_0 = 2000$ m, $v_r = -15$ m/s, and $v_a = 0$ m/s. Without considering the sea clutter and azimuth velocity, the estimated $\hat{f}_{\eta_0}$ is 1147.53 Hz, which is very close to the true value 1147.45 Hz. Then, the estimated result -14.9992 m/s of the radial velocity can be obtained. The results show that the estimation accuracy is very high.

Next, two influencing factors of azimuth velocity and sea clutter are considered, and the accuracy of radial velocity estimation is evaluated by root mean square error (RMSE) through Monte Carlo simulations. Assuming that the number of Monte Carlo simulations is L, the RMSE is calculated as

$$\sigma_{v_r} = \sqrt{\frac{1}{L}\sum_{i=1}^{L}\left(\hat{v}_{r,i} - v_r\right)^2} \tag{43}$$

where $\hat{v}_{r,i}$ is the estimation result of the ith experiment.

Simulate the echo of a moving target whose $x_0 = 2000$ m, $v_r = -15$ m/s. Under each sea clutter model, the SCR varies from 5 to 25 dB. Under each SCR, $L = 100$, and the

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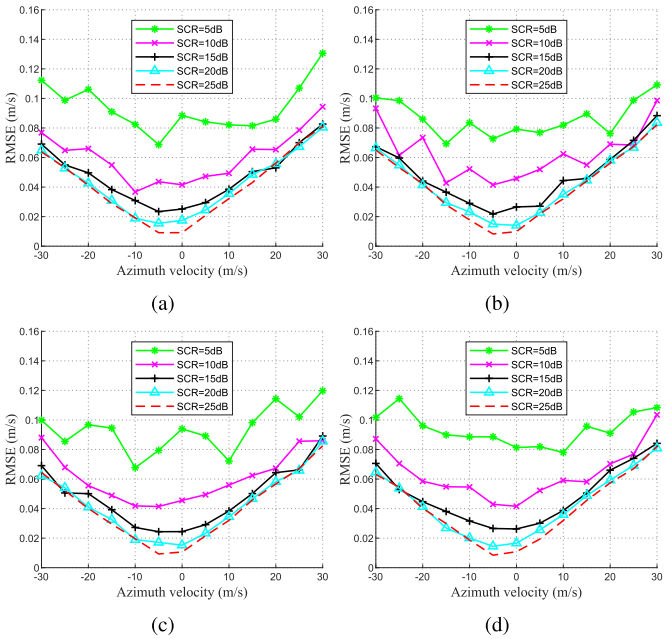


Fig. 12. Influence of clutter and azimuth velocity on radial velocity estimation. The clutter model obeys the distribution of (a) Rayleigh, (b) Weibull, (c) Gama, and (d) K.

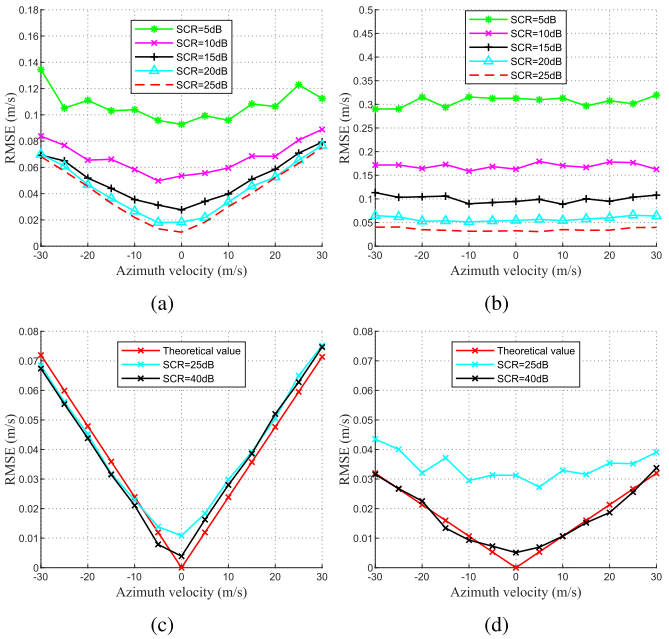


Fig. 13. RMSE of the radial velocity estimated (a) by the proposed method and (b) by the MFCM [29]. Comparison of the experimental and theoretical results of (c) proposed method and (d) MFCM.

TABLE III
RMSE OF THE RADIAL VELOCITY WITH K-DISTRIBUTION CLUTTER MODEL

SCR	v_a=0m/s	v_a=10m/s	v_a=15m/s	v_a=20m/s	v_a=30m/s
5 dB	0.081	0.078	0.096	0.091	0.11
10 dB	0.042	0.059	0.058	0.070	0.103
15 dB	0.026	0.039	0.051	0.066	0.084
20 dB	0.017	0.036	0.049	0.059	0.081
25 dB	0.011	0.032	0.045	0.057	0.082

influence of azimuth velocity on radial velocity estimation is evaluated. The SCR is measured in the coarse-focused image domain. The RMSEs of the radial velocity with four clutter models under different SCRs are shown in Fig. 12(a)–(d). The results show that the estimation error has an approximate linear relationship with azimuth velocity, which is consistent with the theoretical result of (42). When SCR is small, the clutter is the main factor causing the estimation error. From the results, when the SCR is greater than 10 dB, the estimation error is less than 0.1 m/s even if the azimuth velocity reaches 30 m/s. For ships on open sea, these conditions can be basically satisfied. The distribution type of sea clutter has little influence on radial velocity estimation when SCR is greater than 10 dB. Table III lists the RMSEs of radial velocity under K-distribution clutter model.

Finally, to verify the advantages of the proposed radial velocity estimation method in efficiency and accuracy, a comparative experiment is conducted with MFCM [29] mentioned in "Introduction." The MFCM uses the correspondence between the RCM curve and Doppler spectrum to obtain the echo data near zero frequency by intercepting a small part in the middle of the RCM curve, so that the azimuth delay between channels can be easily compensated. The vector $\Gamma(v_r)$ related to the radial velocity can be obtained by frequency correlation

$$\Gamma(v_r) = \text{diag}\left[1, \exp\{j\Delta\}, \ldots, \exp\{j(N-1)\Delta\}\right]$$
$$\Delta = 2\pi T_d \cdot \frac{2v_r}{\lambda} \tag{44}$$

where $T_d = D/2V_{\text{eq}}$ is the time delay between adjacent EPCs, and V_{eq} is the equivalent velocity of the platform, which is equal to $V_s - v_a$. Hence, the estimation result of radial velocity will also introduce an error of $(1-\beta)v_r$ if the azimuth velocity is unknown. The error is proportional to v_a, and the slope is v_r/V_s. The motion of the target will cause the center of the RCM curve not corresponding to the zero Doppler frequency. For low-speed moving targets, a part of the baseband azimuth spectrum (unambiguous azimuth spectrum) can be extracted by truncating a small part in the middle of the RCM curve to estimate the velocity, while for high-speed targets, the baseband azimuth spectrum cannot be obtained. Hence, we set the radial velocity of the moving target as -8 m/s, and $x_0 = 2000$ m. Besides, the azimuth velocity varies from -30 to 30 m/s, and the noise with Rayleigh distribution is added.

The RMSE of our method and MFCM under five different SCRs is shown in Fig. 13(a) and (b). From the results, the accuracy of the proposed method is better than that of the MFCM when SCR is less than 20 dB. Even if the SCR is greater than 20 dB, the accuracy of our method is better when the azimuth velocity is less than 15 m/s. Compared with our method, the estimation error of MFCM is greatly affected by clutter, and the influence of azimuth velocity on radial velocity estimation seems not obvious. Set the SCR to 40 dB, Fig. 13(c) and (d) shows the RMSE results of our method and MFCM,

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TABLE IV
MOTION PARAMETERS OF THE TARGETS

Targets	ST	MT-1	MT-2	MT-3
v_r (m/s)	0	16	-8.5	0
v_a (m/s)	0	0	5	5
x_0 (m)	1600	-500	1600	-1600
f_{η_0}(Hz)	917.96	-286.86	917.35	-917.35

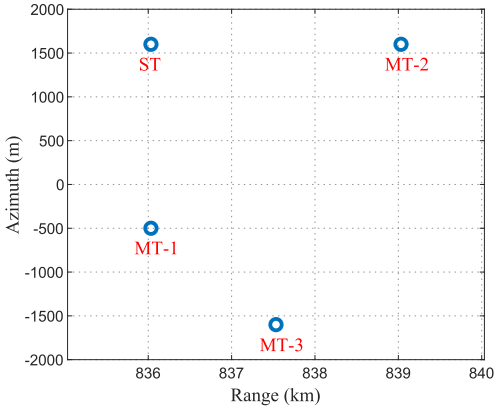


Fig. 14. Imaging scene of the targets.

and compare them with the theoretical values. The results are consistent with the conclusions of theoretical analysis.

In addition, we recorded the running time from extracting the moving target data to estimating the radial velocity with the hardware: AMD Ryzen 2600X Six-Core Processor. The radial velocities of 100 targets are estimated without parallel operations. In our method, the running time required to estimate v_{r_b} and M_{dc} is 126 and 928 s, respectively. The running time required for MFCM is 371 s. The range of v_{r_b} is determined by PRF and wavelength. When PRF $= 1500$ Hz, the maximum v_{r_b} under three typical carrier frequencies ($f_0 = 5.4$, 9.6, and 13.7 GHz) is 20.8, 11.7, and 8.2 m/s, respectively. For the UFS mode of GaoFen-3 SAR, the maximum v_{r_b} is close to 25 m/s. In many cases, the ambiguity number M_{dc} does not need to be estimated. According to the preliminary results, it is proven that the proposed velocity estimation method has advantages in efficiency and accuracy.

B. Simulation Data Experiment

In this part, the unambiguous imaging method of moving target is verified by the simulation data of four channels. The SAR system parameters are listed in Table II. Three moving targets (MT-1, MT-2, and MT-3) and an ST are set in the scene, and the noise with Rayleigh distribution is added. Table IV gives the motion parameters of targets. The imaging scene of targets is shown in Fig. 14. The shortest slant range of MT-1 is the same as that of ST, but its azimuth position is different. The azimuth position of MT-2 is the same as that of ST, but its shortest slant ranges are different. Furthermore, to investigate the influence of azimuth velocity on signal reconstruction, MT-3 is set to move along the azimuth.

TABLE V
AASR AND MAXIMUM P_m OF THE TARGETS

Targets (dB)	ST	MT-1	MT-2	MT-3
$\mathrm{P_{max,org}}$	0.00	-9.99	-11.66	-2.95
$\mathrm{P_{max,aft}}$	0.00	-0.08	0.00	-0.03
$\mathrm{AASR_{org}}$	-42.65	-16.28	-19.44	-42.36
$\mathrm{AASR_{aft}}$	-42.89	-42.33	-43.24	-42.31

($\mathrm{P_{max,org/aft}}$ is the maximum power of real target. Index "org" denotes the result without considering motion parameters. "aft" denotes the result processed by the proposed method.)

Fig. 15(a) shows the imaging result of the reconstructed signal without considering motion parameters. From the result, ST and MT-3 are reconstructed correctly, while MT-1 and MT-2 are reconstructed incorrectly, resulting in false targets along the azimuth. The yellow ellipse and the red box mark the focused position of the false target and real target, respectively. The successful reconstruction of MT-3 indicates that the influence of azimuth velocity on signal reconstruction can be ignored. By comparing the focused position of ST and MT-2, the result shows that the azimuth-focused position of the moving target deviates from the correct position because the radial velocity is not considered. The blue solid circle indicates the correct focused positions of moving targets. MT-1 and ST are located in the same range gate, and their azimuth profile is presented in Fig. 15(b). The result shows that incorrect reconstruction of MT-1 not only leads to serious SNR loss of the real target but also increases the amplitude of false targets. The maximum amplitude of false targets is -16.28 dB relative to the real target. The azimuth profile of MT-2 is plotted in Fig. 15(c). Since the azimuth velocity is not considered in imaging, resulting in the azimuth defocus and aggravating the SNR loss. Table V lists AASR and maximum P_m of four targets.

Process the simulation data by the flow of Fig. 2. First, range compression is carried out on the echoes of all the channels. Fig. 16(a) shows the data structure of the reference channel signal in the R-D domain, and the enlarged view of MT-2 is on the right. It can be seen that for any frequency cell, the Doppler spectrum is scattered in four positions, and there is an offset of the spectrum center. Since ST and MT-1 have the same shortest slant range, their echoes cannot be separated in the R-D domain. After azimuth deramp processing, the data structure of the reference channel signal in the deramp frequency domain is shown in Fig. 16(b), and the enlarged view of MT-2 is on the right. It can be seen that the Doppler spectrum is compressed into several azimuth frequency cells, and the targets are coarsely focused. ST and MT-1 are focused on different azimuth cells so that their echoes can be separated. It should be mentioned that the result shown in Fig. 16 does not add the noise for the convenience of observation.

The targets are detected in the deramp frequency domain and their echoes are extracted from the coarse-focused SAR images. The parameter f_{η_0} of the targets is estimated, and its baseband radial velocities can be calculated according to (16). Fig. 17 shows the estimated results of the parameter f_{η_0}. Then, the reference channel signal is used to estimate the ambiguity

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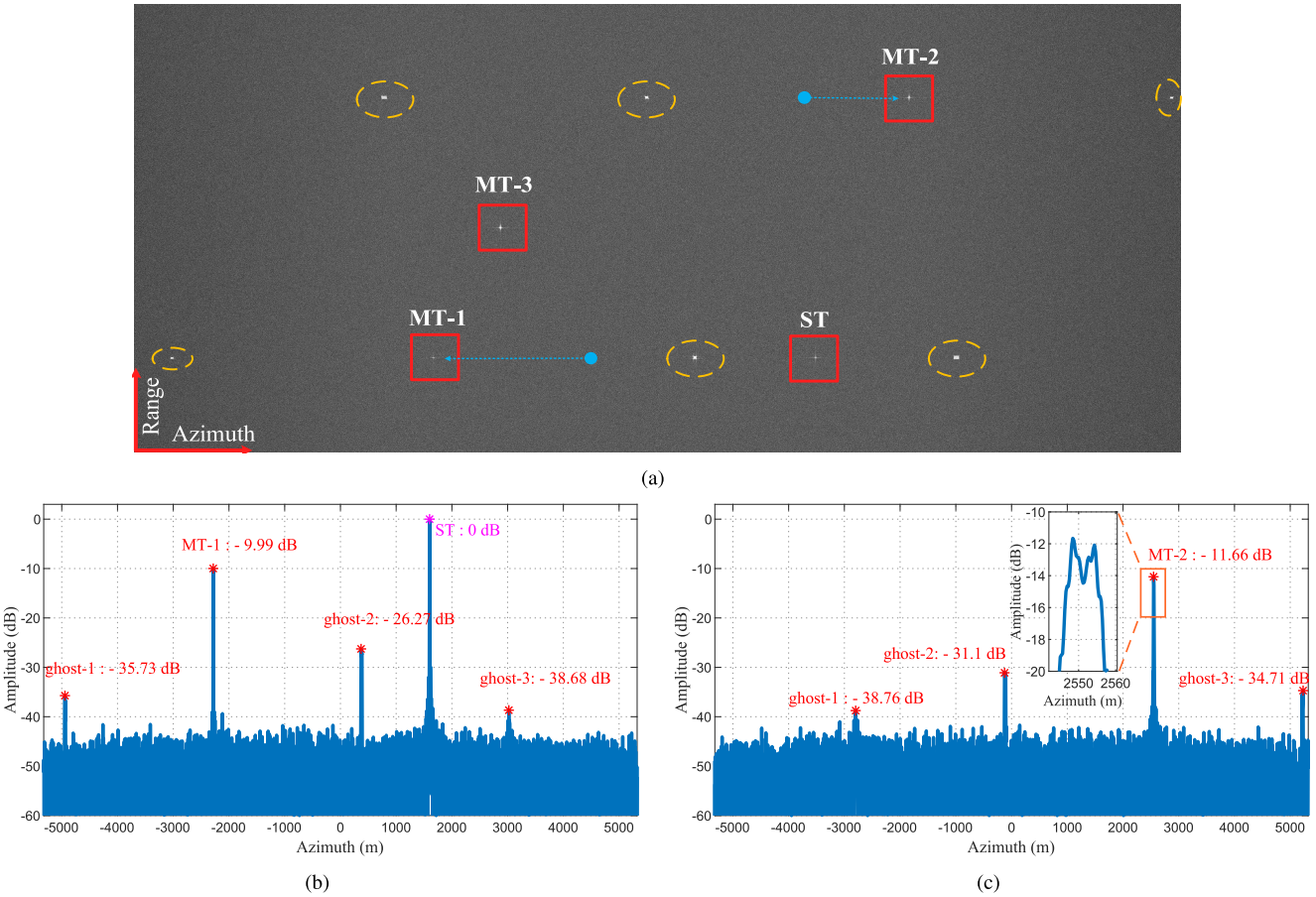


Fig. 15. Imaging result of the reconstructed signal without considering the motion parameters. (a) SAR image. (b) Azimuth profile of ST and MT-1. (c) Azimuth profile of MT-2.

TABLE VI
ESTIMATION RESULT OF THE RADIAL VELOCITY

Targets	SCR	True value	Our method	MFCM[29]
ST	19.91 dB	0.00 m/s	0.002 m/s	-0.054 m/s
MT-1	17.87 dB	16.00 m/s	15.97 m/s	16.233 m/s
MT-2	16.97 dB	-8.50 m/s	-8.508 m/s	-8.536 m/s
MT-3	20.00 dB	0.00 m/s	0.019 m/s	0.048 m/s
Ship-1	13.85 dB	3.41 m/s	2.87 m/s	2.78 m/s
Ship-2	13.28 dB	2.36 m/s	2.43 m/s	2.18 m/s
Ship-3	12.24 dB	2.13 m/s	2.04 m/s	2.29 m/s
Ship-4	20.65 dB	unknown	3.58 m/s	3.64 m/s
Ship-5	21.49 dB	unknown	1.44 m/s	1.35 m/s

number M_{dc}. Taking MT-1 as an example, Fig. 18(a) successively presents the coarse-focused result of MT-1 when M_a takes different values. The results show that the focused result is optimal when $M_a = 1$. For comparison, the radial velocity is also estimated using MFCM, and the estimation results of both the methods are all listed in Table VI. For MFCM, since the radial velocity of MT-1 is relatively large, only a very small part of the intercepted RCM curve corresponds to the unambiguous azimuth spectrum. The radial velocity estimated at different azimuth frequency points jumps from near 16 m/s to near -7.8 m/s, with a difference of $\lambda \mathrm{PRF}/2$. In addition, only a small part of azimuth frequency range corresponds to the radial velocity near 16 m/s. The final velocity is obtained by manually intercepting these values and taking their average value, but there is still a large error. The estimation result of our method is better than that of MFCM, and the estimation error is less than 0.02 m/s.

Next, according to the estimation result of radial velocity and f_{η_0}, the signal of all the targets are reconstructed by the flow of Fig. 6. To obtain the fine focused results of MT-2 and MT-3, the azimuth velocity is considered into the imaging process, i.e., the platform velocity V_s is replaced by the equivalent velocity $V_s - v_a$. Therefore, each target is imaged separately here. Finally, the imaging results of these targets are superimposed onto the imaging result of the background. The result is shown in Fig. 19(a). Similarly, Fig. 19(b) gives the azimuth profile of MT-1 and ST. It can be seen that false targets of MT-1 are suppressed below -40 dB, and MT-1 is focused on correct azimuth position without SNR loss. Fig. 19(c) gives the azimuth profile of MT-2, and the false targets are also suppressed. To facilitate comparison, AASR and maximum P_m of four targets are also listed in Table V.

C. Real Data Experiment of GaoFen-3 SAR

GaoFen-3 is the first Chinese multichannel SAR satellite. Its UFS mode adopts dual-receiving technique to obtain

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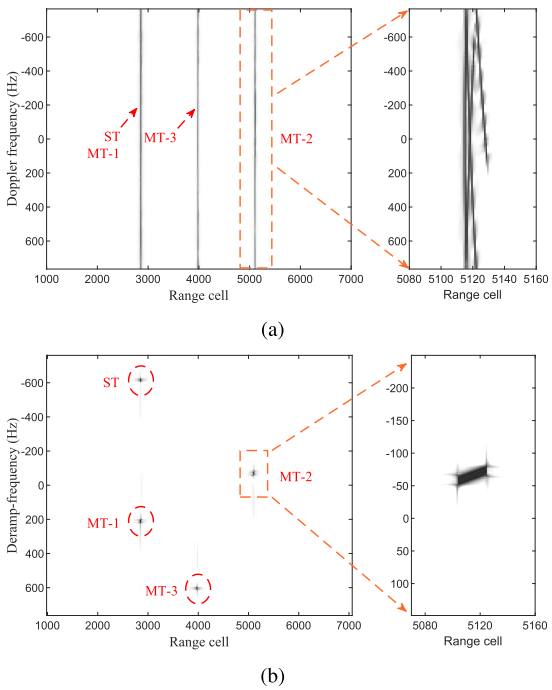


Fig. 16. Data structure of the reference channel echo (a) in the R-D domain and (b) in the deramp frequency domain.

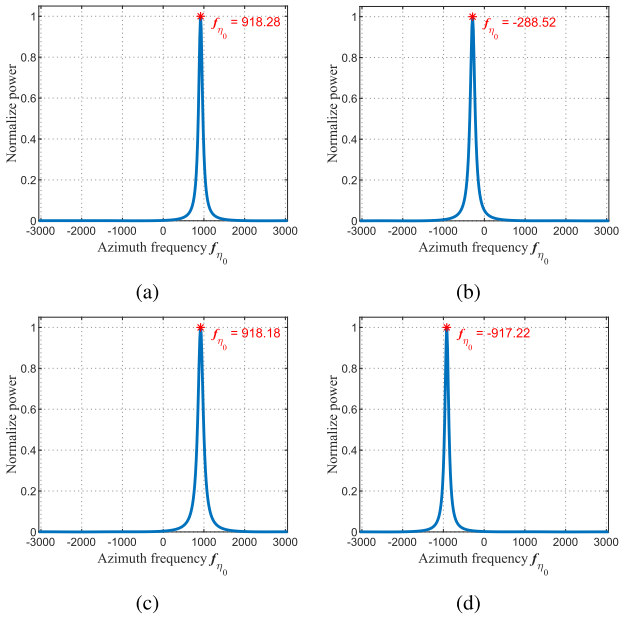


Fig. 17. Estimation result of the parameter $f_{\eta 0}$ for the four targets. (a) ST. (b) MT-1. (c) MT-2. (d) MT-3.

3-m resolution and 30-km swath simultaneously [40]. Two maritime scenes containing multiple ships in UFS mode are selected to verify the proposed algorithm. The system parameters and image information are listed in Table VII. The strip width (on the slant range plane) of the first scene is 10 km, in which there are three ships and the sea conditions are more complicated. Processing the real data with the flow of Fig. 2. First, range compression and azimuth deramp processing are carried out on real data, and we can obtain two coarse-focused SAR images, one of which is shown in Fig. 20.

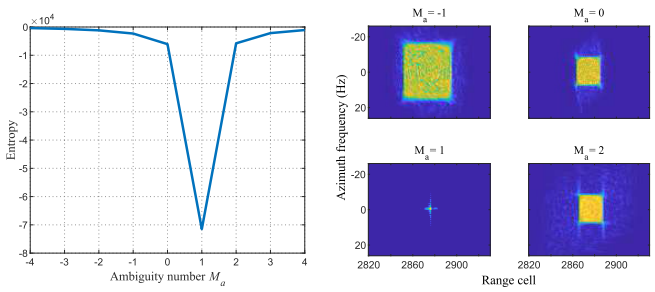


Fig. 18. Estimation result of the parameter M_{dc} for MT-1. (a) Coarse-focused MT-1 when M_a takes different values. (b) Entropy of the coarse-focused MT-1.

TABLE VII
SYSTEM PARAMETERS OF GAOFEN-3 SAR IN UFS MODE

Parameters	Value
Platform Velocity	7563 m/s
Carrier frequency	5.4 GHz
Pluse duration	50 μs
Antenna length	7.5 m
Number of channels	2
Distance between channels	3.75 m
$\mathrm{PRF_s}$ of the first scene	1884 Hz
Incidence angle of the first scene	22.37°
$\mathrm{PRF_s}$ of the second scene	1996.2 Hz
Incidence angle of the second scene	36.88°

($\mathrm{PRF_s}$ denotes the PRF for each channel.)

The coarse-focused ships have higher energy than sea clutter. Then, the ship positions are detected by the GLRT detector, and the echoes of the ships are extracted from coarse-focused SAR images. The radial velocities of ships are estimated by the proposed method, which are listed in Table VI.

Next, the signals of all the ships are reconstructed by the flow of Fig. 6. Since the purpose of the real data experiment is to verify the unambiguous imaging method, the azimuth velocity is not considered in the imaging process, so all the ships can be imaged together. Finally, the imaging result of ships is superimposed onto the imaging result of the sea clutter. For comparison, Fig. 21(a) shows the imaging result of the reconstructed signal without considering motion parameters. There are obvious false targets above ships, whereas the other false targets are submerged in the strong sea clutter. The red arrow indicates real ships and the yellow box marks false targets. The imaging result of the reconstructed signal processed by the proposed method is shown in Fig. 21(b). From the result, false targets are suppressed so that they are submerged in the sea clutter and cannot be recognized. Ships are focused on the position of the ship's wake.

The AIS can provide dynamic information of the ship such as SOG, course over ground (COG), and navigation position (latitude and longitude). We provided the AIS information service organization with SAR imaging time, ship's longitude and latitude information, and the longitude and latitude coordinates of the area where Ship-1, Ship-2, and Ship-3 are located. Based on these information, they obtained the AIS information of ships that coincided with the imaging time through the

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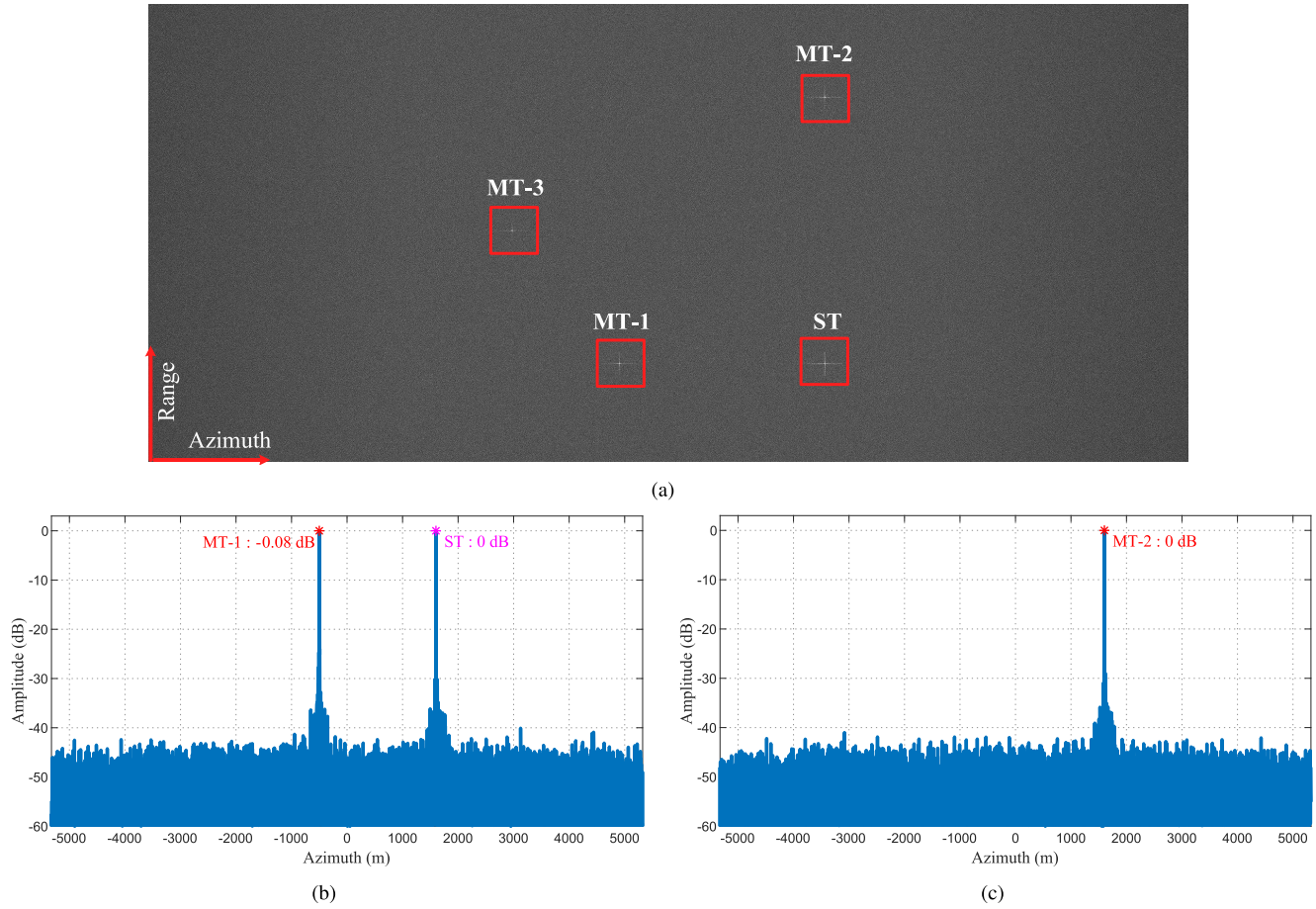


Fig. 19. Imaging result processed by the proposed method. (a) SAR image. (b) Azimuth profile of ST and MT-1. (c) Azimuth profile of MT-2.

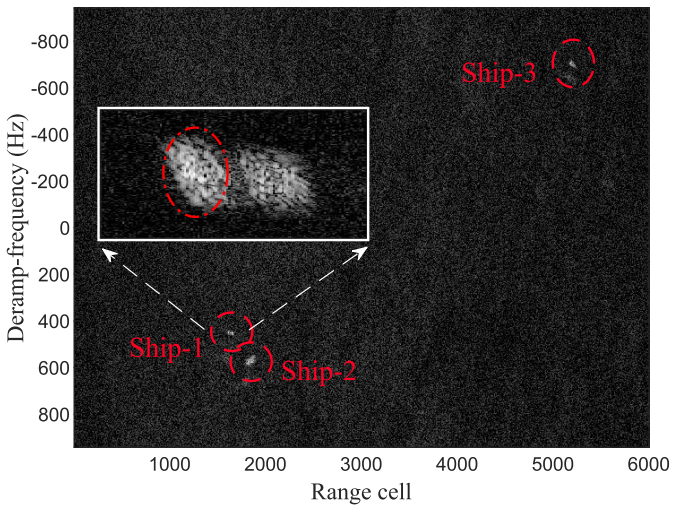


Fig. 20. Coarse-focused SAR image of the first scene.

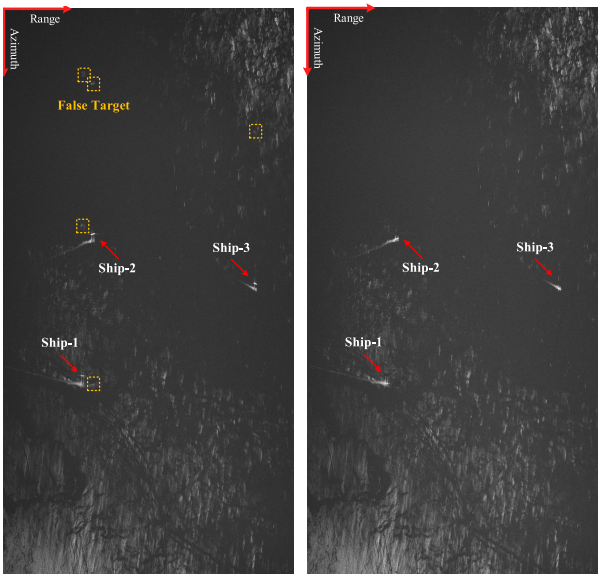
TABLE VIII
AIS INFORMATION OF SHIPS

Ship ID	Longitude	Latitude	COG	SOG
Ship-1	120°17.841′ E	35°2.866′ N	69.9°	17.6 knots
Ship-2	120°18.888′ E	35°0.138′ N	86.9°	12.3 knots
Ship-3	120°25.377′ E	35°2.228′ N	56.3°	11.7 knots

ship trajectory playback function. The key information are listed in Table VIII. According to the flight direction (347.97°) of the platform, the ship's COG, and the incidence angle (22.37°), the radial velocities of Ship-1, Ship-2, and Ship-3 can be calculated as 3.41, 2.36, and 2.13 m/s, respectively. Compared with the estimation result of our method, the errors of Ship-2 and Ship-3 are 0.07 and 0.09 m/s, respectively. Ship-1 has a large estimation error of 0.54 m/s, which may be caused by the fact that Ship-1 is divided into two parts with high SCR and low SCR, as shown in the white box of Fig. 20. We reestimated the radial velocity with higher SCR data (marked by red ellipse), and the result is 3.29 m/s. The error is reduced to 0.12 m/s. (The original estimate is used in signal reconstruction.) In addition, the radial velocity is also estimated using MFCM, the estimation errors of Ship-1, Ship-2, and Ship-3 are 0.63, 0.18, and 0.16 m/s, respectively. Ship-1 also has a large estimation error.

Fig. 22(a) shows the amplitude diagrams of the real target and false targets of Ship-2 in Fig. 21(a). The real ship deviates from the wake, and the false target is obvious. Fig. 21(b) shows the amplitude diagrams of the real target and false targets of Ship-2 in Fig. 21(b). Fig. 22(c) shows the azimuth profile of imaging result of Ship-2 (taking the average of the 20 range

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(a) (b)

Fig. 21. Imaging result of the reconstructed signal: (a) without considering the motion parameters and (b) that processed by the proposed method.

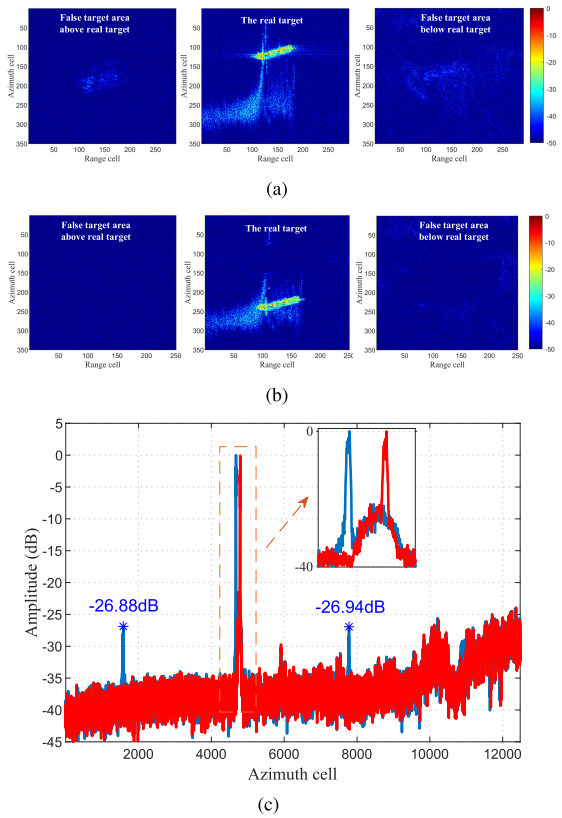


Fig. 22. Imaging results of the real target and false targets of Ship-2: (a) without considering motion parameters and (b) that processed by the proposed method. (c) Azimuth profile of the imaging result of Ship-2; the blue line and red line are the results of not considering and considering motion parameters, respectively.

bins where the ship is located), where the blue line is the result without considering the motion parameters, and the red line is the result processed by the proposed method. From the result of Fig. 22(b) and (c), the real ship is repositioned

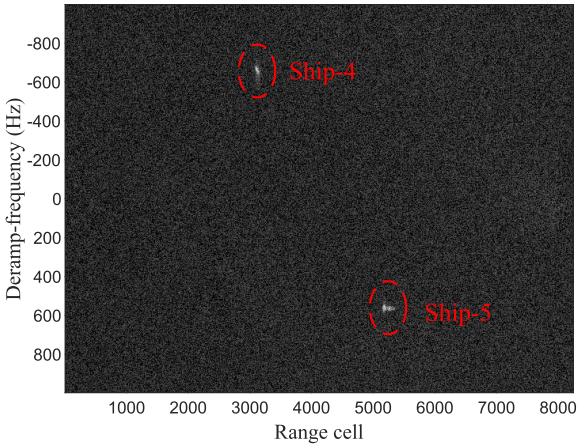


Fig. 23. Coarse-focused SAR image of the second scene.

TABLE IX
ACR RATIOS OF THE SHIPS

Ship ID	Ship-1	Ship-2	Ship-3	Ship-4	Ship-5
$\mathrm{ACR_{up,org}}$ (dB)	10.53	10.28	9.84	*	9.26
$\mathrm{ACR_{up,aft}}$ (dB)	*	*	*	*	*
$\mathrm{ACR_{down,org}}$ (dB)	*	9.79	*	3.71	8.24
$\mathrm{ACR_{down,aft}}$ (dB)	*	*	*	*	*

(The symbol "*" is used to mark the false targets that submerged in the sea clutter, and their ACR values are not calculated.)

to the position of the wake, and false targets are completely suppressed. The experimental results prove that the proposed signal reconstruction method has the advantage that the error of radial velocity does not lead to false targets.

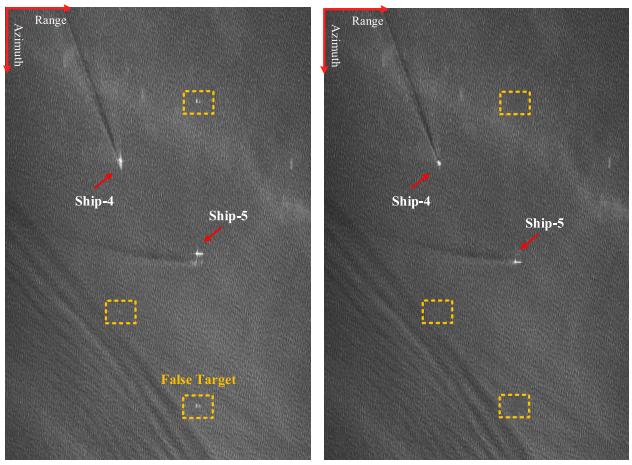
To quantitatively evaluate the suppression effect of false targets, we define an evaluation metric

$$\mathrm{ACR} = 10 \cdot \log 10 \left\{ \frac{P_{\mathrm{amb,ave}}}{P_{\mathrm{clu,ave}}} \right\} \tag{45}$$

where $P_{\mathrm{amb,ave}}$ is the average power of the false target. $P_{\mathrm{clu,ave}}$ is the average power of the sea clutter. The better the effect of false target suppression, the smaller the ratio of ACR. The symbols $\mathrm{ACR_{up,org/aft}}$ and $\mathrm{ACR_{down,org/aft}}$ are used to represent the ACR value of the first false target above and below the real target, respectively. Subscript "org" represents the result without considering motion parameters. Subscripts "aft" represents the result processed by the proposed method. The ACR values of false targets for Ship-1, Ship-2, and Ship-3 are listed in Table IX.

In the second scene, the swath width on the slant range plane is 11.5 km, and the sea conditions are relatively simple. Processing the real data with the flow of Fig. 2. The coarse-focused SAR image is shown in Fig. 23, in which there are two ships. Fig. 24(a) shows the imaging result of the reconstructed signal without considering the motion parameters. The red arrow indicates real ships and the yellow box marks false targets. Since the radial velocity of Ship-4 is small, the energy of its false target is low and the false target is not obvious in the image. The imaging result of the reconstructed signal processed by the proposed method is shown in Fig. 24(b).

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(a) (b)

Fig. 24. Imaging result of the reconstructed signal: (a) without considering the motion parameters and (b) that processed by the proposed method.

Since the SCR of Ship-4 and Ship-5 is higher than 20 dB, the radial velocity estimation results of the two methods are relatively close, as shown in Table VI. In addition to suppressing false targets, the ship can also be accurately located. The ACR values of false targets for Ship-4 and Ship-5 are listed in Table IX.

VI. Conclusion

Unambiguous imaging of moving targets is an important task of HRWS SAR technology, especially for moving ships in ocean scenes. The inaccuracy of radial velocity and channel imbalance are the main factors leading to the degradation of signal reconstruction performance. To improve the signal reconstruction performance of moving target and suppress false targets in SAR images, an unambiguous imaging scheme based on azimuth deramp technology is proposed. To facilitate the extraction of moving target echoes, azimuth deramp processing is carried out on the echoes of all the channels to obtain coarse-focused ship. The GLRT algorithm is used to detect moving targets, and their echoes are extracted from the sea clutter. According to the characteristics of the moving target signal in the deramp frequency domain, the radial velocity is divided into the baseband part and the ambiguity part. The estimation of the baseband part is transformed into spatial spectrum estimation, and the ambiguity part can be estimated by minimizing the entropy of the coarse-focused target. This method does not need iterative operation and has high efficiency and accuracy. In addition, it does not need redundant channels and can be used to fast-moving targets. The proposed signal reconstruction method uses the characteristics of the coarse-focused signal and performs zero filling of the signal (whose azimuth spectrum is compressed) in the deramp frequency domain. The zero-filled signal is modulated by a new azimuth FM function whose azimuth time sampling interval is $1/M_{\text{hrws}}$ times of the original signal, so as to obtain the up-sampling signal in the time domain, and then obtain the reconstructed signal. The signal reconstruction performance of this method is not affected by channel imbalance, and false targets can be removed even if there is radial velocity error. Clutter suppression is not considered in this scheme. For the targets that may exist on the sea, such as islands, reefs, and ships, the estimated radial velocity can be used to judge whether they are moving targets. For nonmoving targets, signal reconstruction and imaging processing are performed together with the sea clutter. In the experimental part of the measured data, the performance of the radial velocity method is verified by the ship's AIS information, and the imaging results prove that the proposed signal reconstruction method has the advantage that radial velocity error does not lead to false targets. The proposed scheme can be used not only for the existing DRC SAR systems but also for future multichannel HRWS SAR systems.

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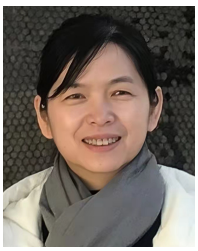
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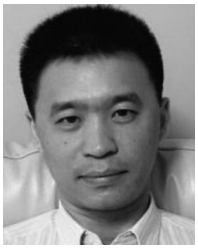
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